DNA Artifact Classification
Machine Learning Model Comparison Report

Report Generated: December 07, 2025

PROJECT OVERVIEW

This report presents a comprehensive comparison of machine learning models for DNA artifact classification in Low Template DNA (LTDNA) analysis.

CLASSIFICATION TASK

- Target Classes: Allele, Back_Stutter, Forward_Stutter, Pull_Up
- Models Evaluated: 9 models
- Resampling Strategies: Original, SMOTE, RandomOverSampler

MODELS ANALYZED

- Extra Trees Classifier
- LightGBM
- XGBoost
- Random Forest
- TabPFN (Tabular Prior-Fitted Networks)
- Support Vector Machine
- Logistic Regression
- Multi-Layer Perceptron
- Gaussian Naive Bayes

EVALUATION METRICS

- Accuracy & Balanced Accuracy
- F1-Score (Macro)
- Mean Squared Error (MSE) & Mean Absolute Error (MAE)
- Log Loss
- Per-Class Precision, Recall, F1-Score
- Confusion Matrices
- ROC & Precision-Recall Curves
- Feature Importance (tree-based models)
- SHAP Values (select models)

Overall Model Performance Comparison

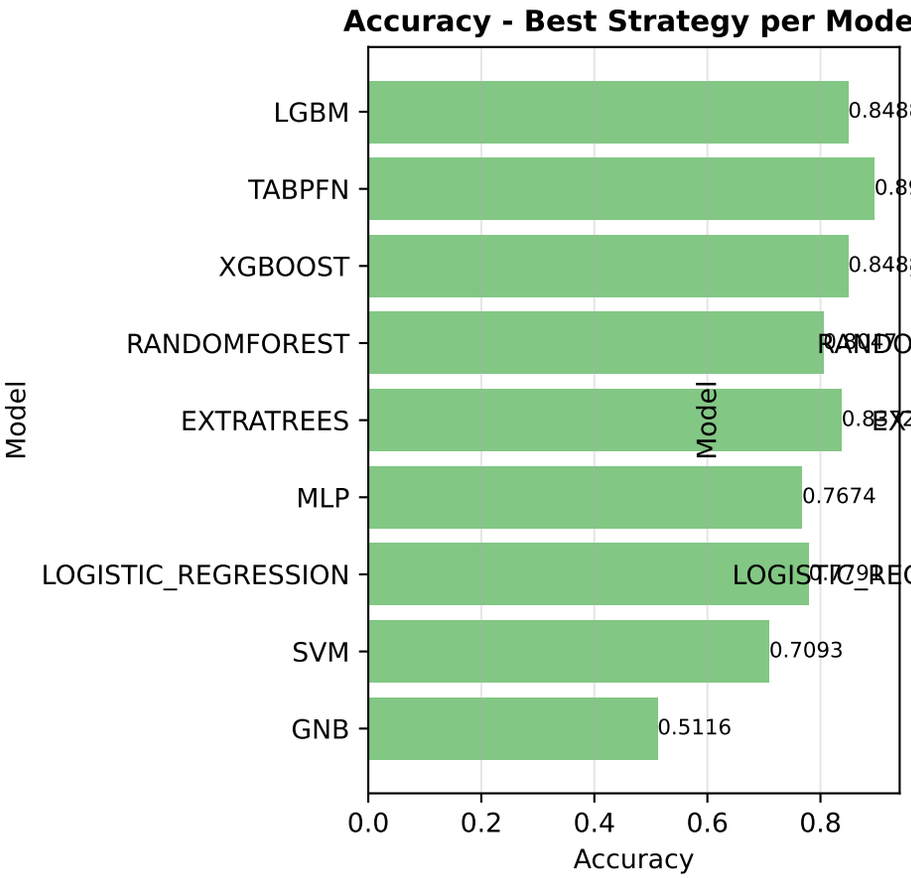
Top 10 Model-Strategy Combinations (by Balanced Accuracy)

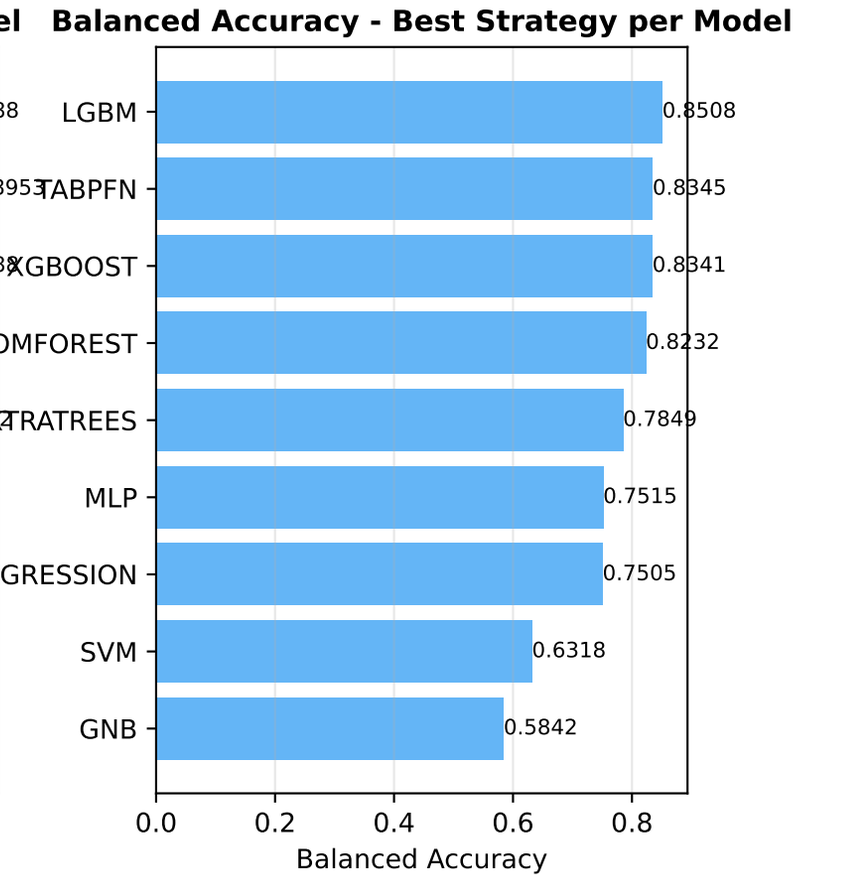
Model	Strategy	Accuracy	Balanced Acc	F1 (macro)	Log Loss
LGBM	RandomOverSampler	0.8488372093023255	0.8507936507936508	0.8368209362017411	0.7490527932744093
TABPFN	Original	0.8953488372093024	0.8345238095238094	0.8508098891730604	0.2778900933371078
XGBOOST	SMOTE	0.8488372093023255	0.8341269841269842	0.8159103298602184	0.4737109970471048
RANDOMFOREST	SMOTE	0.8046875	0.8231688778330569	0.7724141005319101	0.5573815216602453
LGBM	Original	0.8488372093023255	0.8123015873015873	0.8130776880776881	0.6926304660789888
RANDOMFOREST	SVM SMOTE	0.78125	0.8119748479823101	0.7640574251480661	0.5640587117446246
LGBM	SMOTE	0.8372093023255814	0.8067460317460318	0.8006425948592411	0.7597141846026401
EXTRATREES	SMOTE	0.8372093023255814	0.7849206349206349	0.7728648170814631	0.4834729516475584
EXTRATREES	Original	0.8372093023255814	0.7547619047619047	0.7519215845058541	0.4643431008707415
MLP	SMOTE	0.7674418604651163	0.7515376984126985	0.7232103854012542	0.9889185075559588

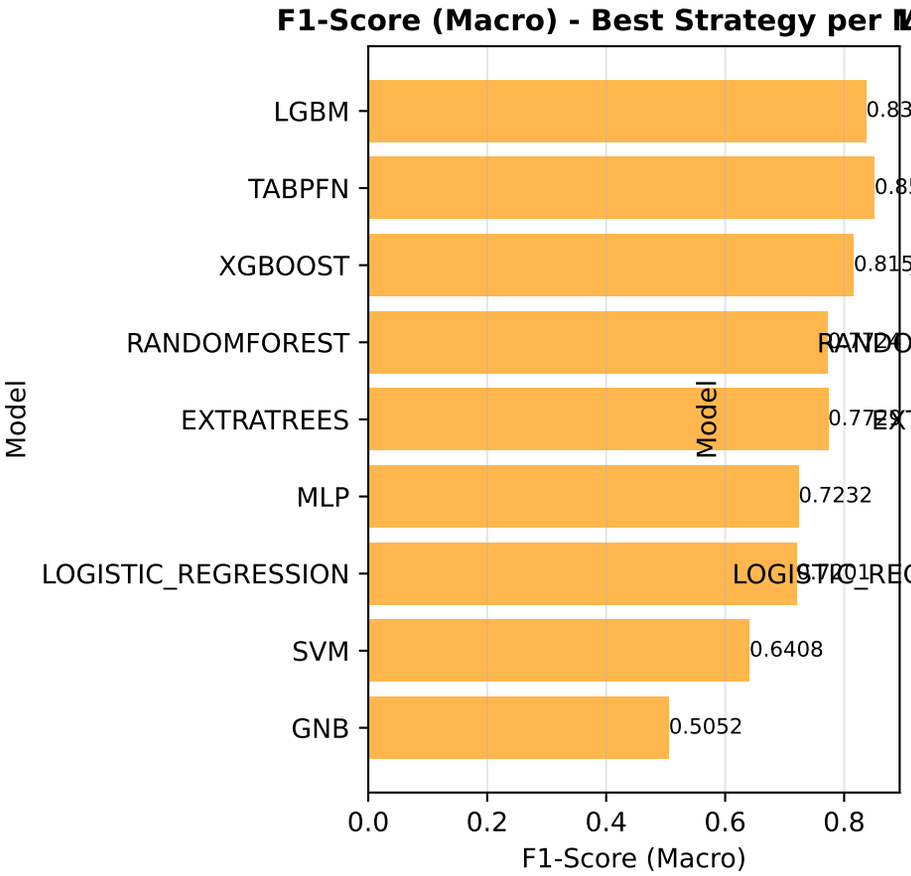
Best Strategy per Model (by Balanced Accuracy)

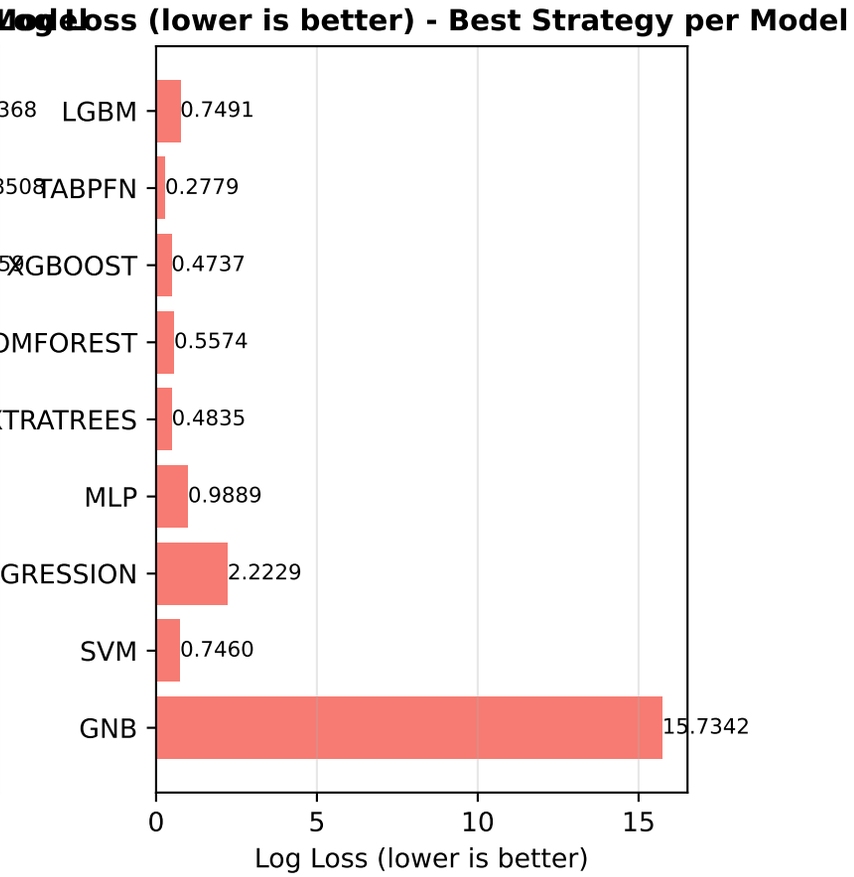
Model	Strategy	Accuracy	Balanced Acc	F1 (macro)	Log Loss
LGBM	RandomOverSampler	0.8488372093023255	0.8507936507936508	0.8368209362017411	0.7490527932744093
TABPFN	Original	0.8953488372093024	0.8345238095238094	0.8508098891730604	0.2778900933371078
XGBOOST	SMOTE	0.8488372093023255	0.8341269841269842	0.8159103298602184	0.4737109970471048
RANDOMFOREST	SMOTE	0.8046875	0.8231688778330569	0.7724141005319101	0.5573815216602453
EXTRATREES	SMOTE	0.8372093023255814	0.7849206349206349	0.7728648170814631	0.4834729516475584
MLP	SMOTE	0.7674418604651163	0.7515376984126985	0.7232103854012542	0.9889185075559588
LOGISTIC_REGRESSION	RandomOverSampler	0.7790697674418605	0.7504960317460317	0.7200538998097652	2.2229453403481028
SVM	SMOTE	0.7093023255813954	0.6317956349206351	0.6407837165408374	0.7460284193316125
GNB	SMOTE	0.5116279069767442	0.5841765873015873	0.5052254958971376	15.73424445540344

Performance Metrics Comparison

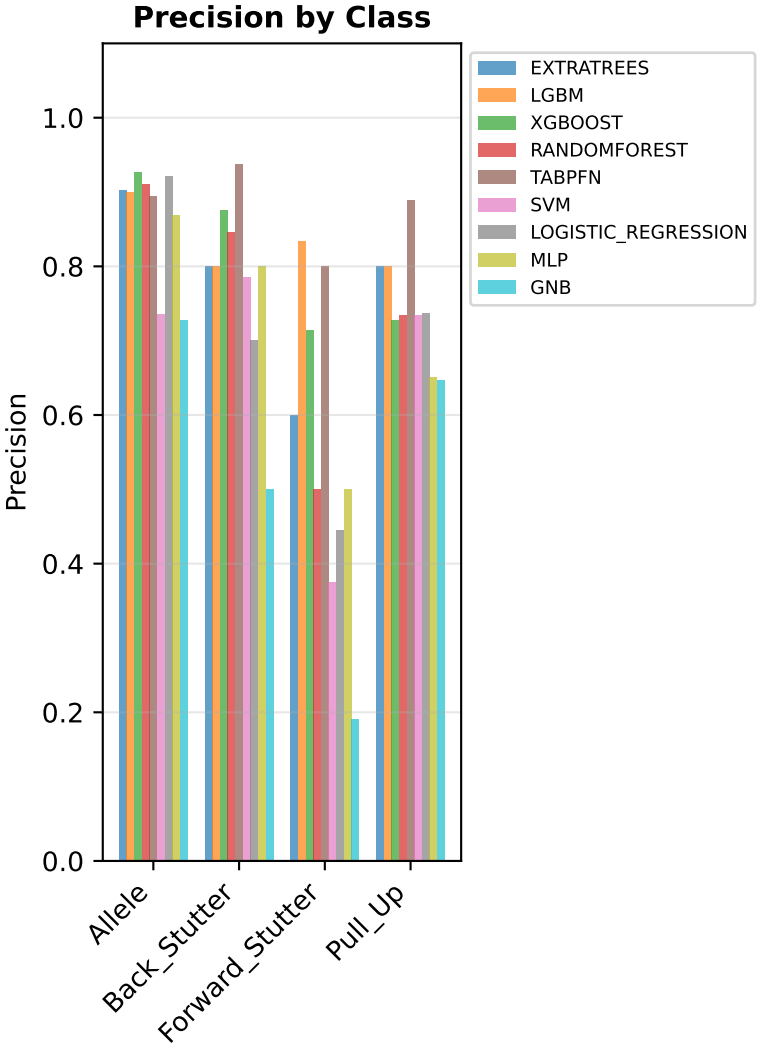


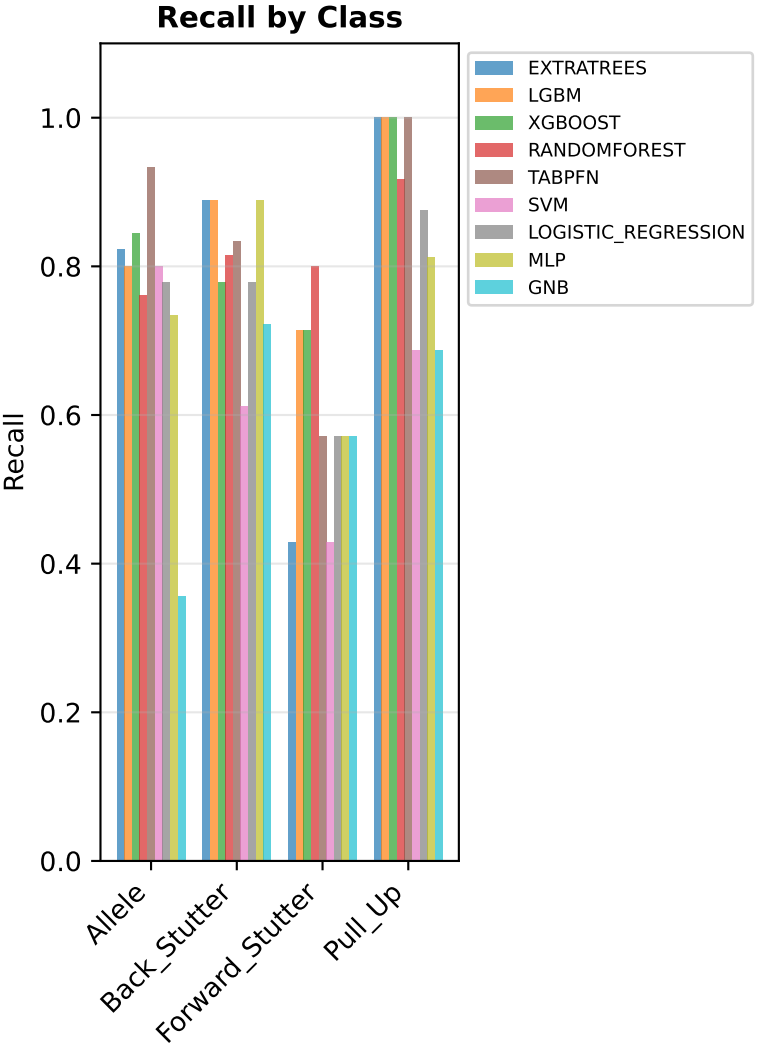


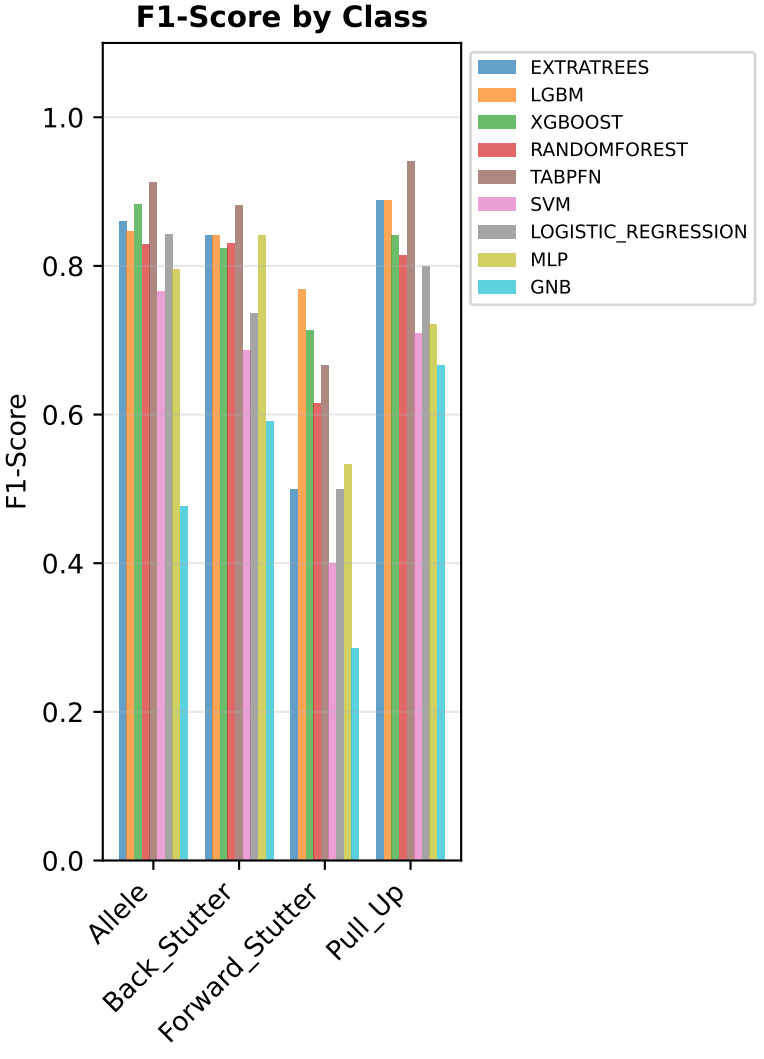


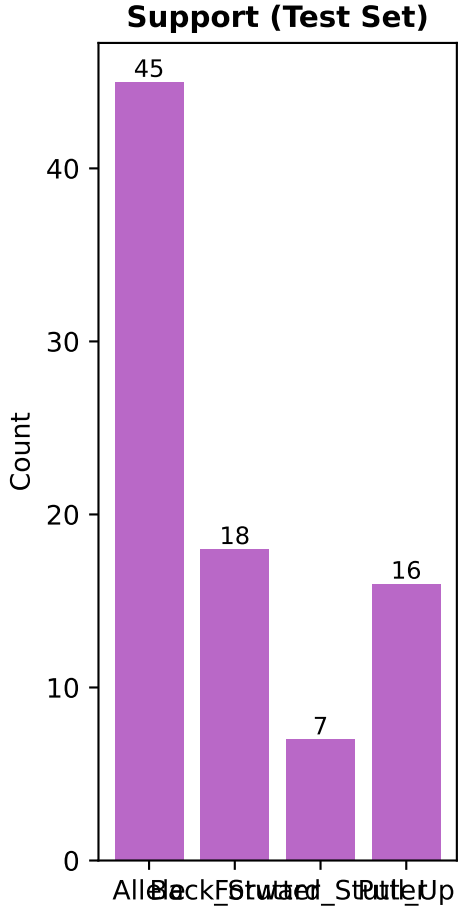


Per-Class Performance Comparison (Best Strategy per Model)







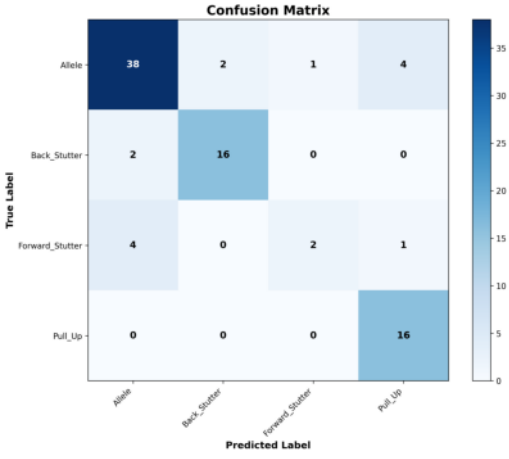


EXTRATREES - Original Strategy

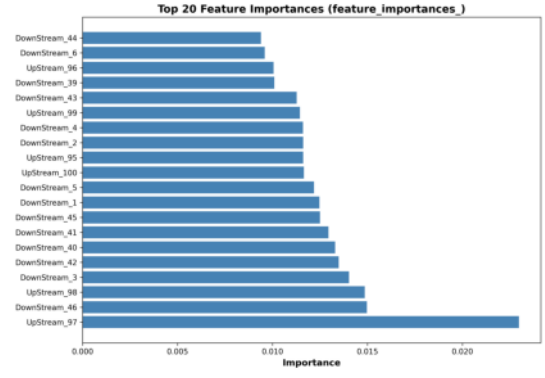
Performance Metrics

Accuracy	0.8372
Balanced Accuracy	0.7548
F1-Score (Macro)	0.7519
MSE	0.7093
MAE	0.3140
Log Loss	0.4643

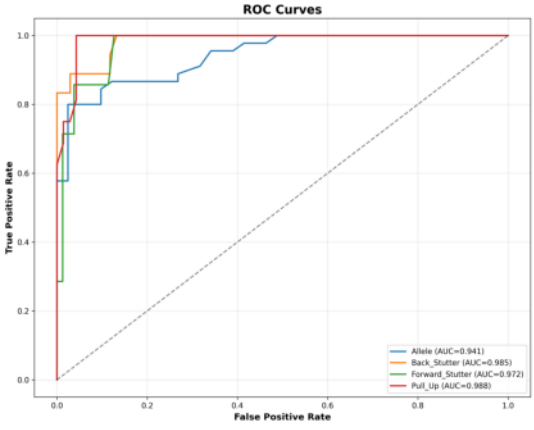
Confusion Matrix



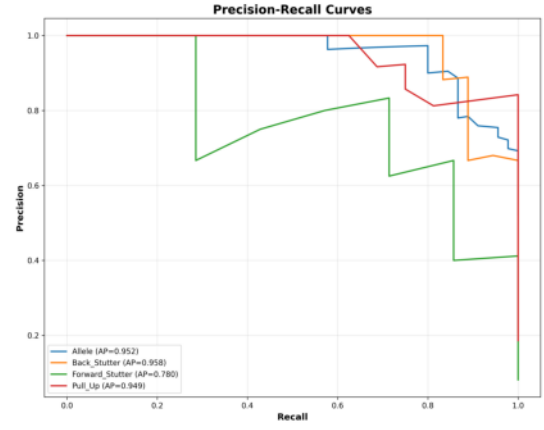
Feature Importance



ROC Curves



Precision-Recall Curves

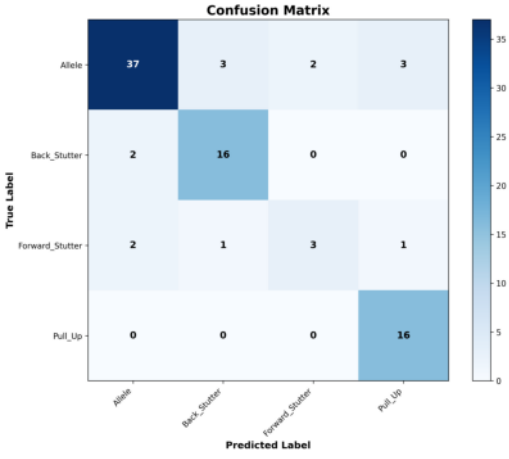


EXTRATREES - SMOTE Strategy

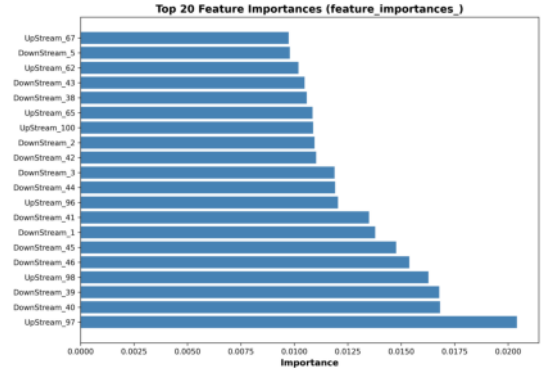
Performance Metrics

Accuracy	0.8372
Balanced Accuracy	0.7849
F1-Score (Macro)	0.7729
MSE	0.5814
MAE	0.2791
Log Loss	0.4835

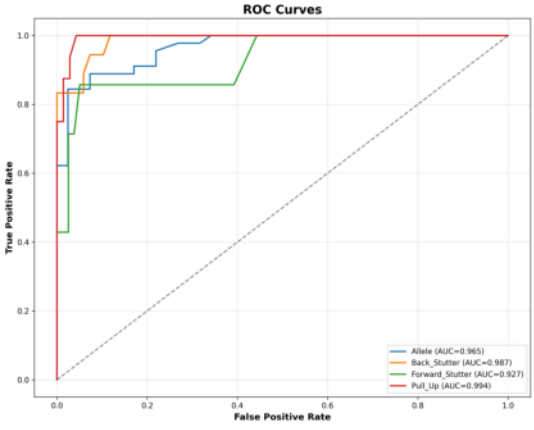
Confusion Matrix



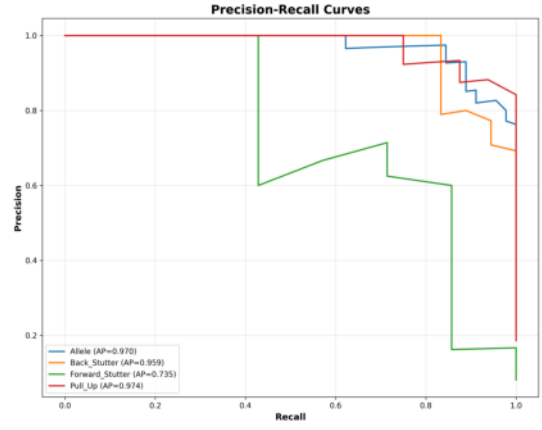
Feature Importance



ROC Curves



Precision-Recall Curves

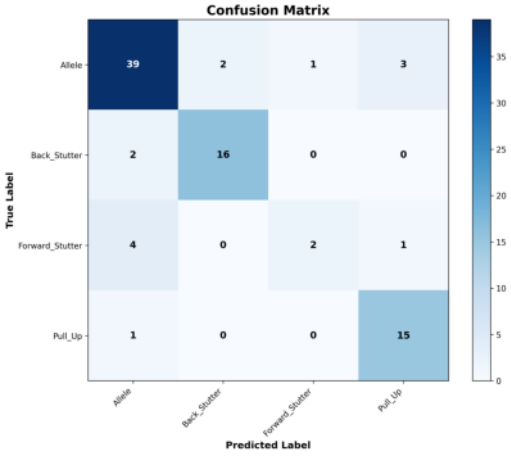


EXTRATREES - RandomOverSampler Strategy

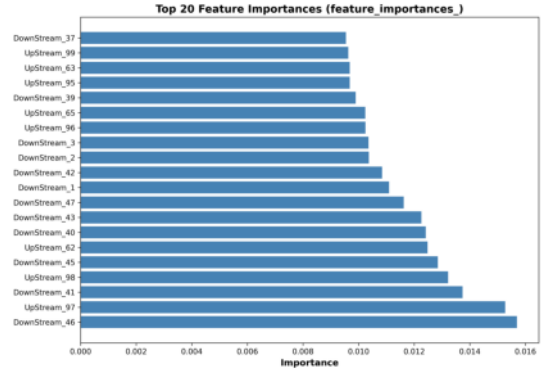
Performance Metrics

Accuracy	0.8372
Balanced Accuracy	0.7447
F1-Score (Macro)	0.7508
MSE	0.7093
MAE	0.3140
Log Loss	0.5079

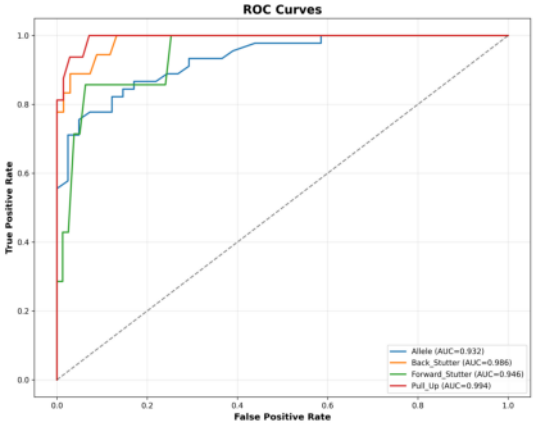
Confusion Matrix



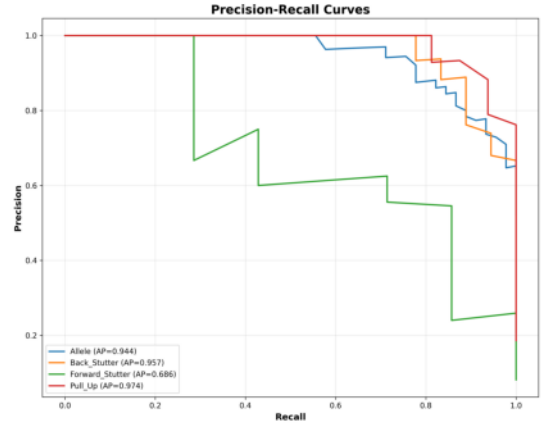
Feature Importance



ROC Curves



Precision-Recall Curves

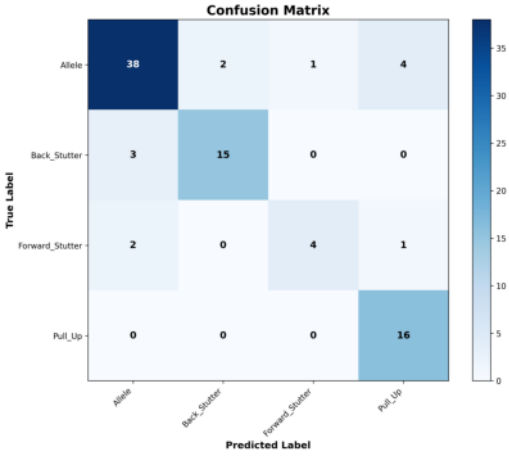


LGBM - Original Strategy

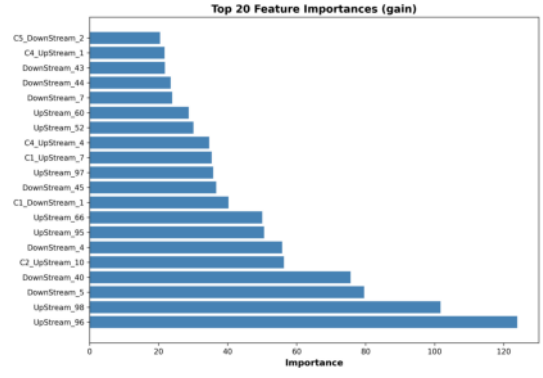
Performance Metrics

Accuracy	0.8488
Balanced Accuracy	0.8123
F1-Score (Macro)	0.8131
MSE	0.6279
MAE	0.2791
Log Loss	0.6926

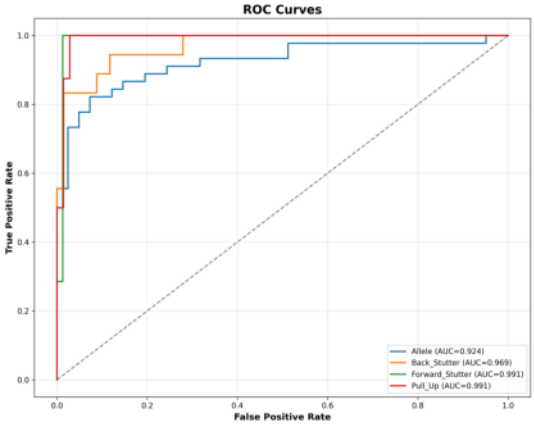
Confusion Matrix



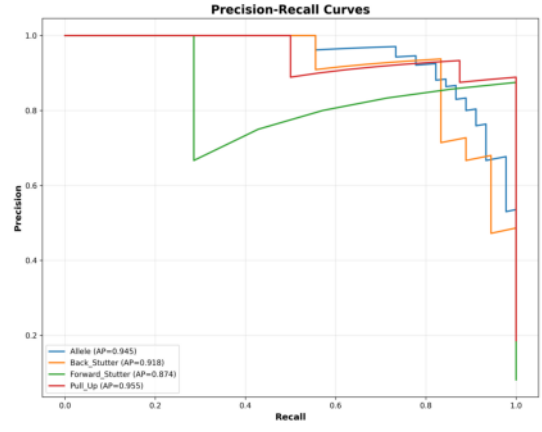
Feature Importance



ROC Curves

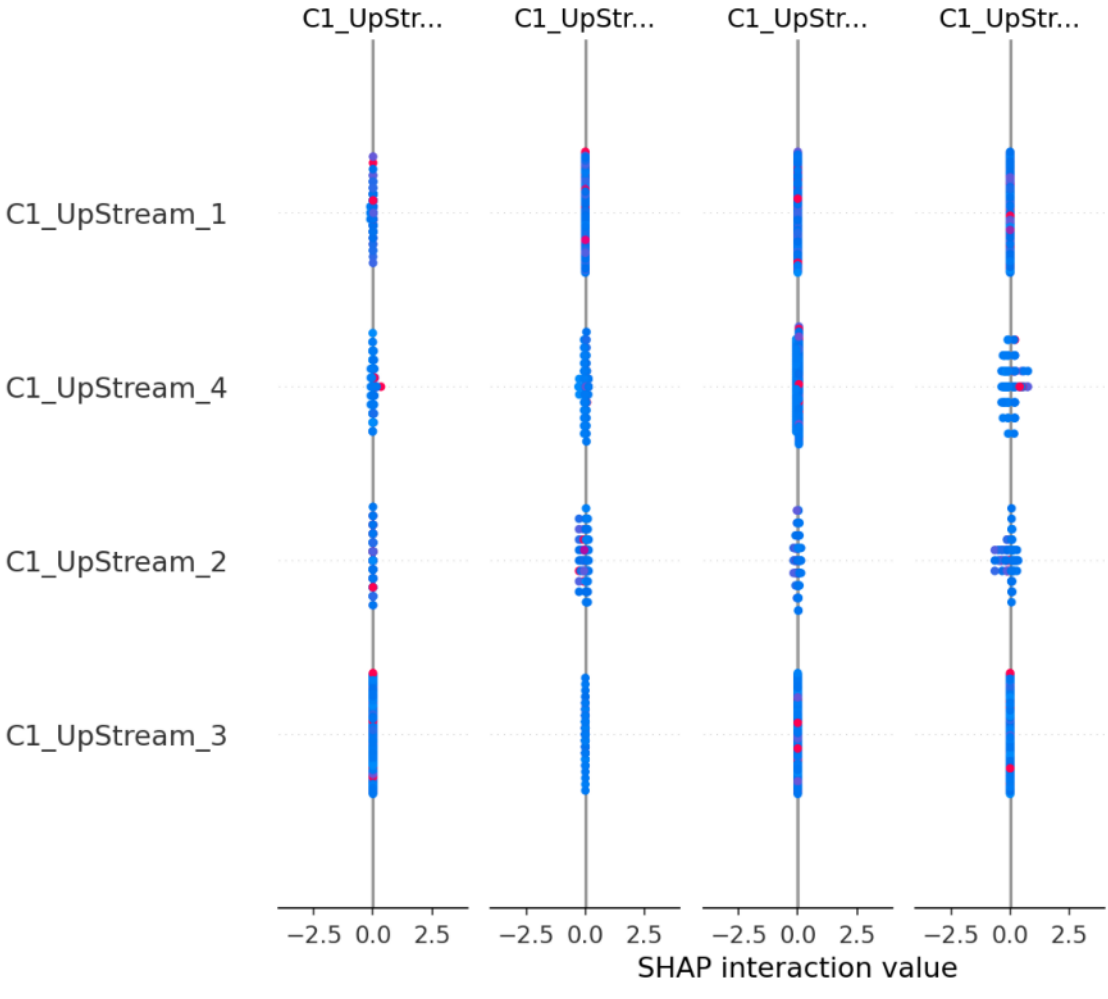


Precision-Recall Curves

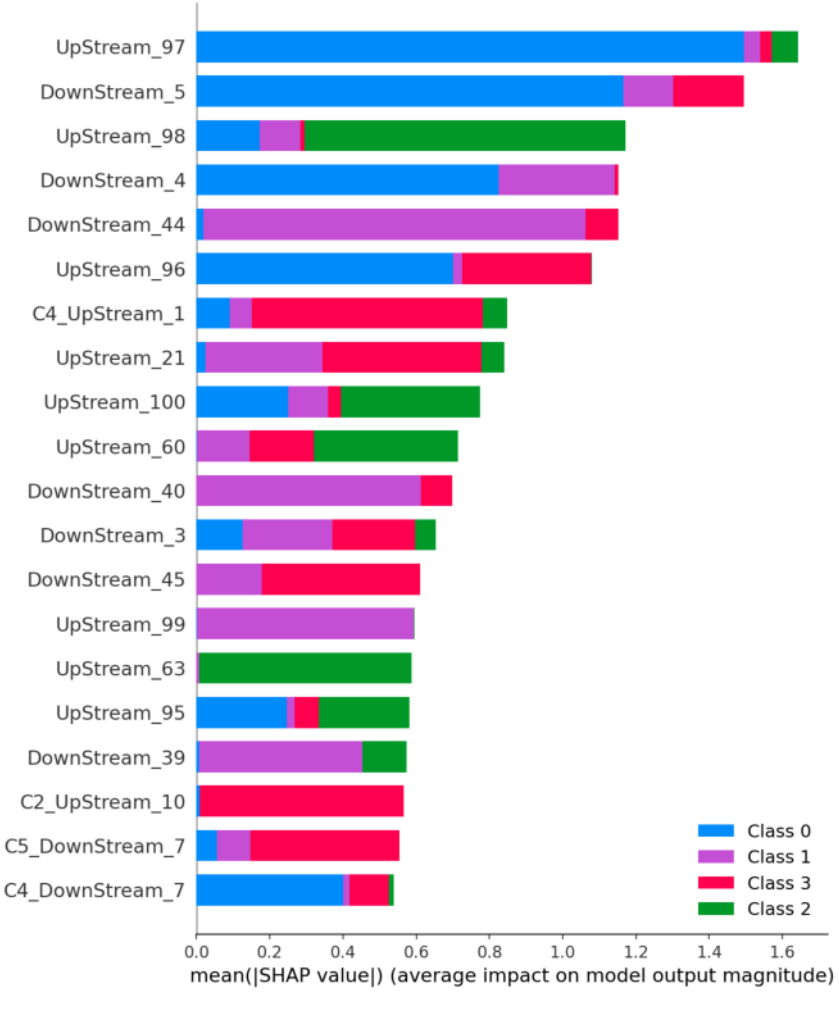


LGBM - Original - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

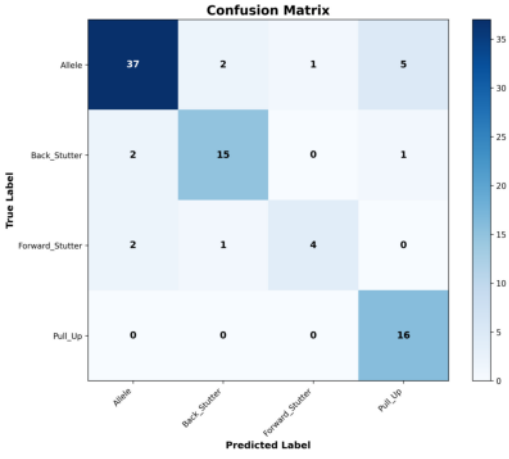


LGBM - SMOTE Strategy

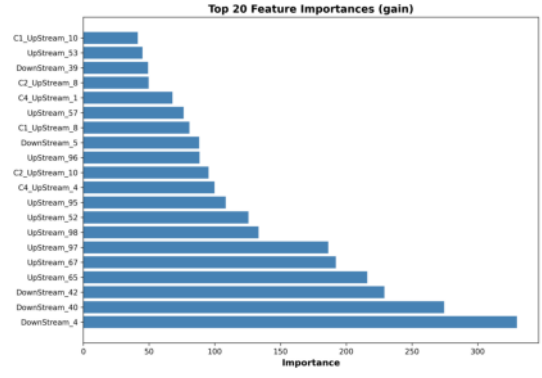
Performance Metrics

Accuracy	0.8372
Balanced Accuracy	0.8067
F1-Score (Macro)	0.8006
MSE	0.7674
MAE	0.3256
Log Loss	0.7597

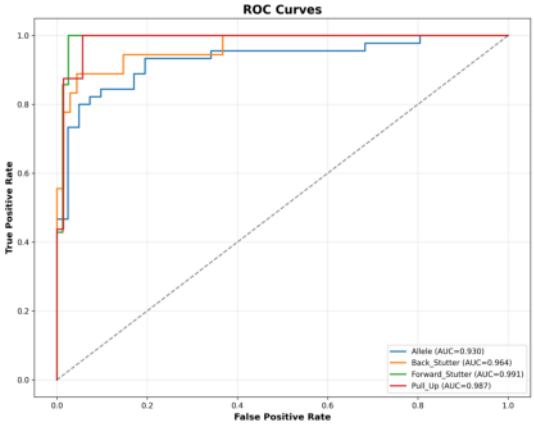
Confusion Matrix



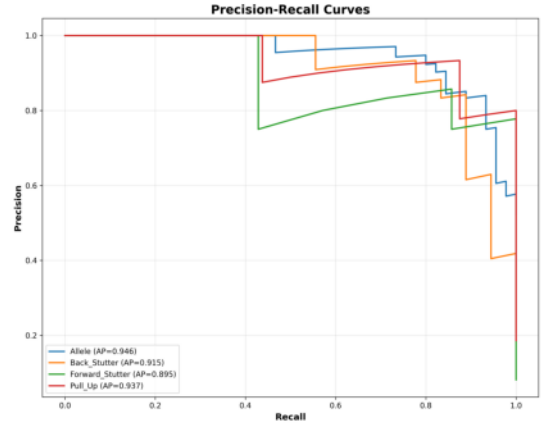
Feature Importance



ROC Curves

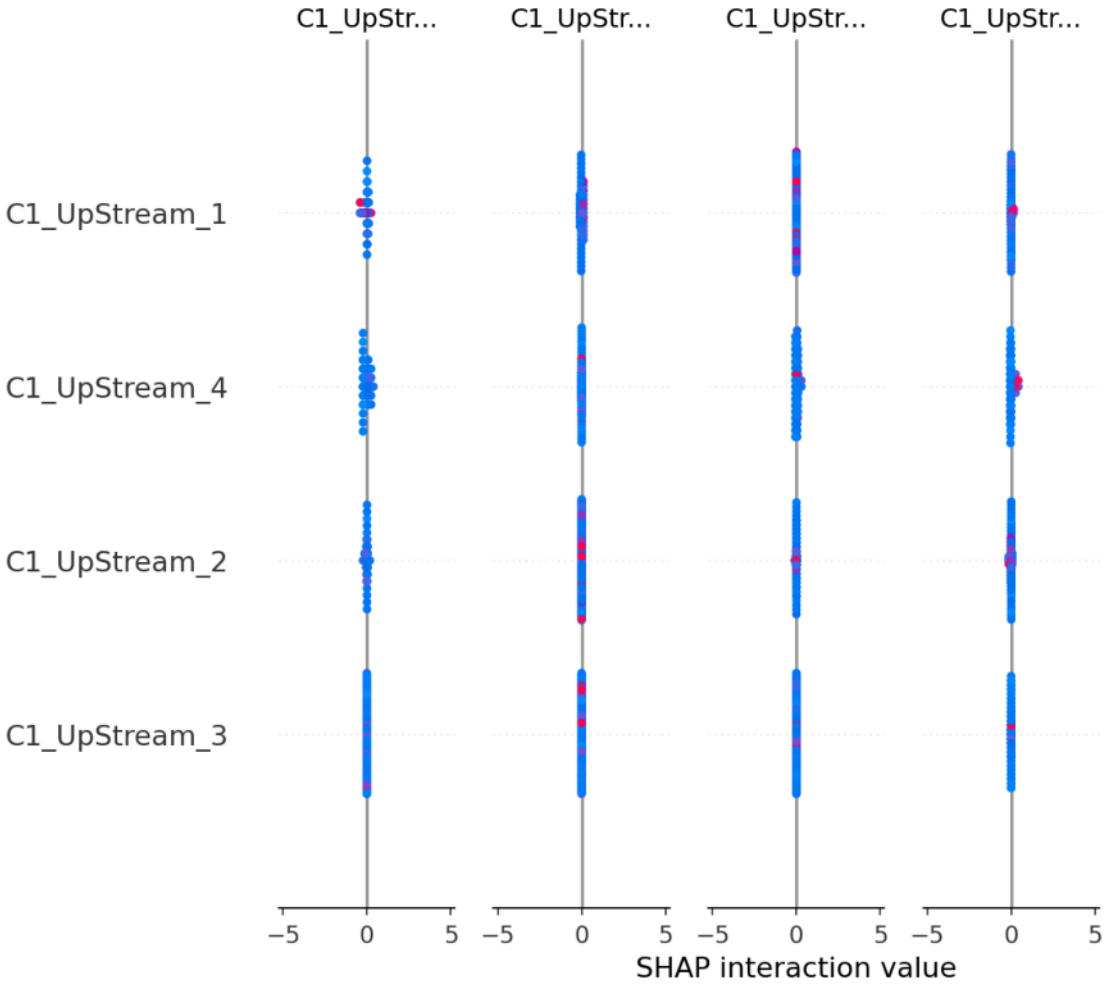


Precision-Recall Curves

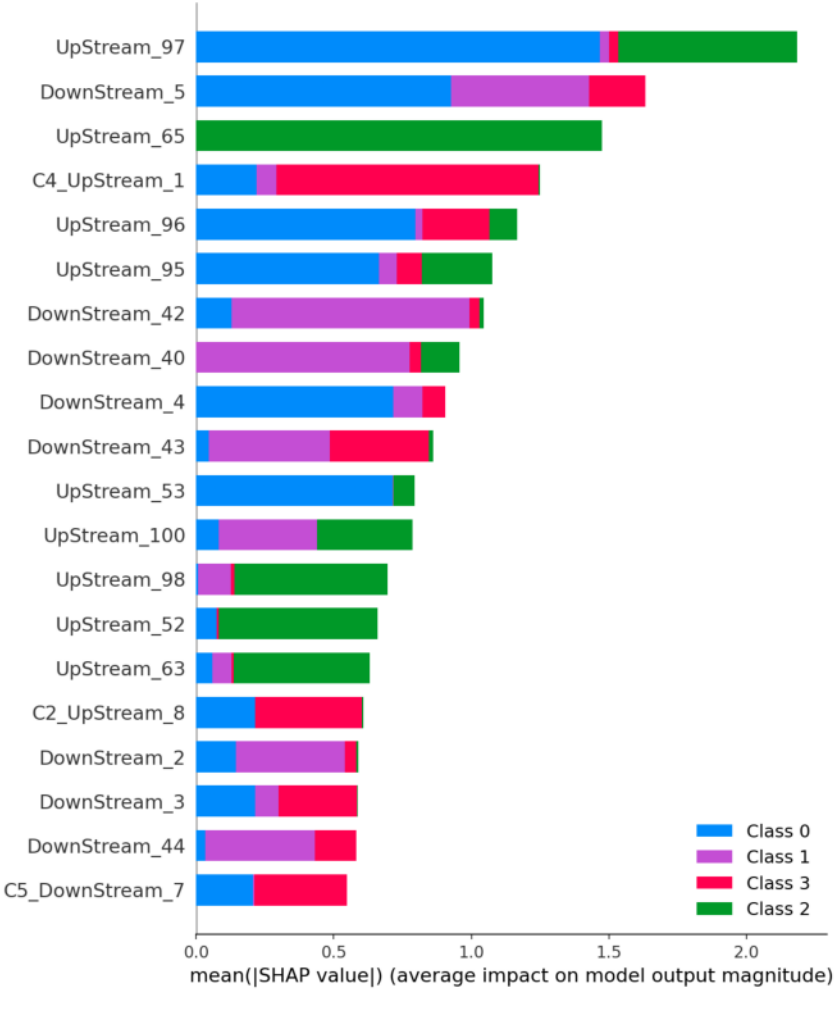


LGBM - SMOTE - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

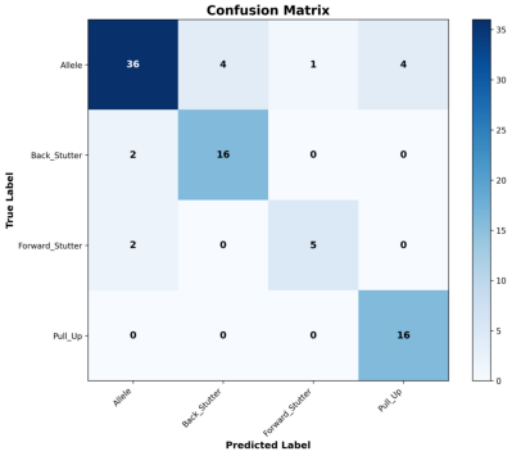


LGBM - RandomOverSampler Strategy

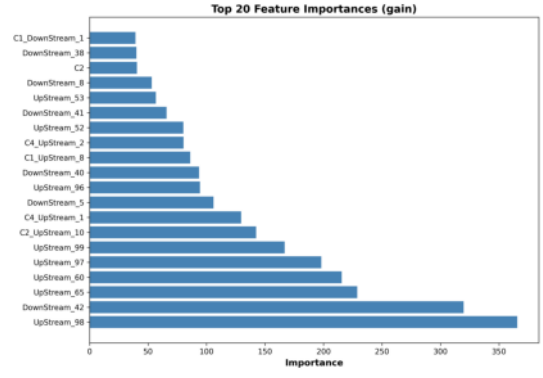
Performance Metrics

Accuracy	0.8488
Balanced Accuracy	0.8508
F1-Score (Macro)	0.8368
MSE	0.6279
MAE	0.2791
Log Loss	0.7491

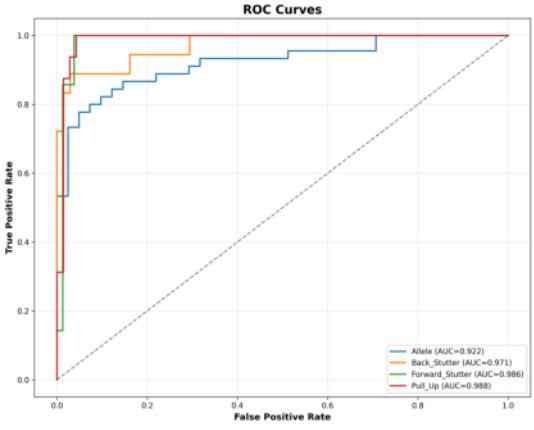
Confusion Matrix



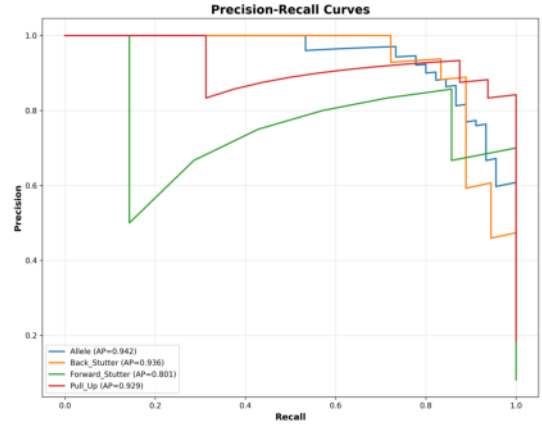
Feature Importance



ROC Curves

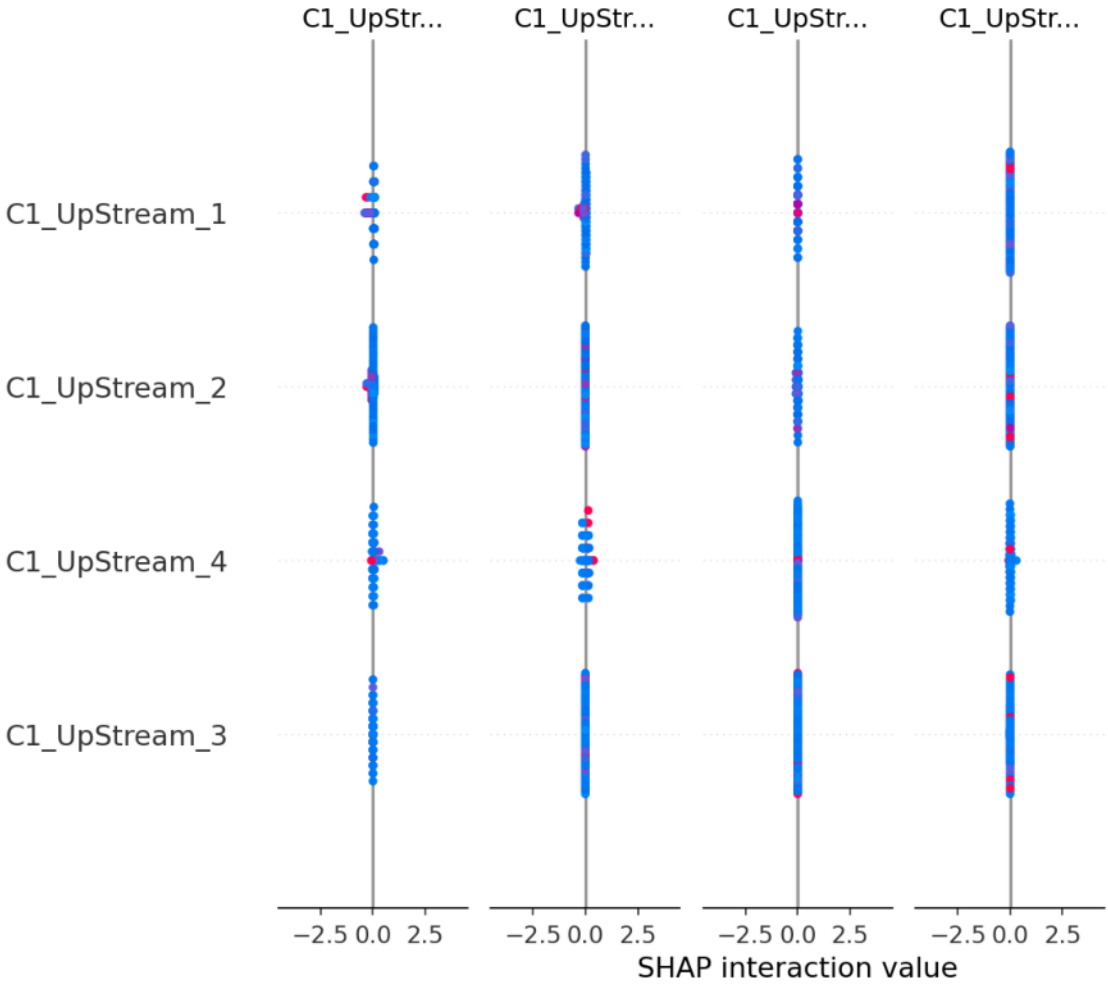


Precision-Recall Curves

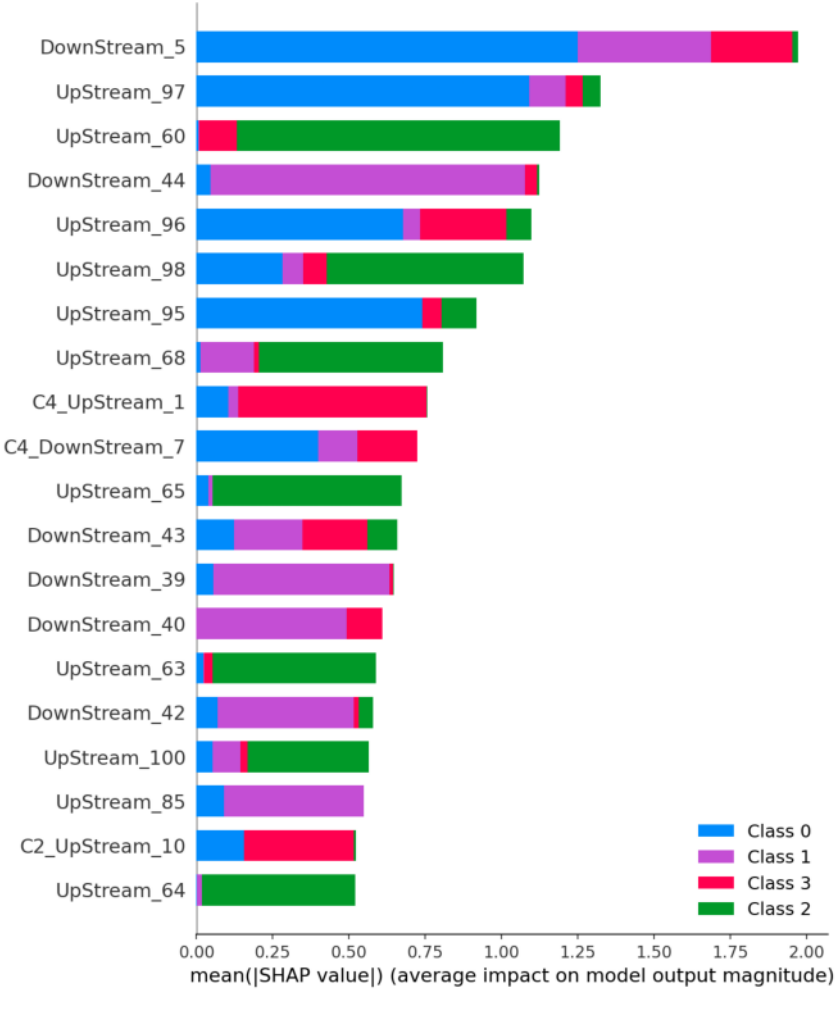


LGBM - RandomOverSampler - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

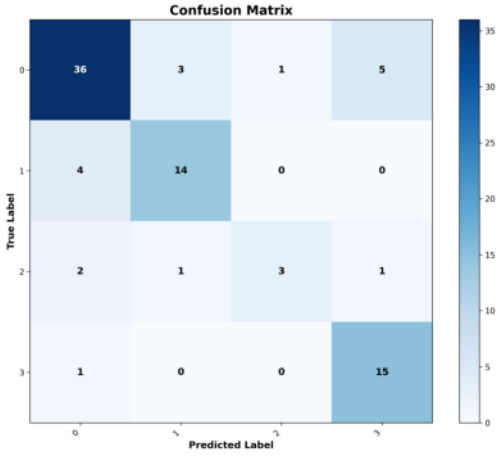


XGBOOST - Original Strategy

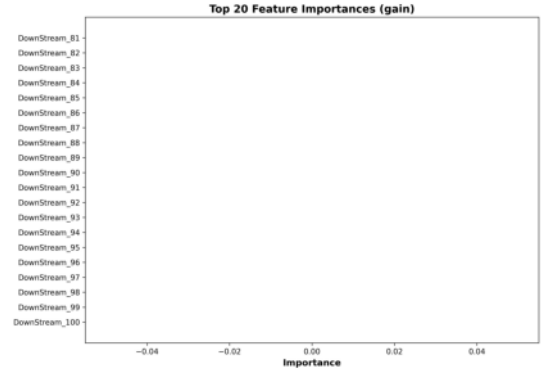
Performance Metrics

Accuracy	0.7907
Balanced Accuracy	0.7360
F1-Score (Macro)	0.7381
MSE	0.8721
MAE	0.3837
Log Loss	0.5129

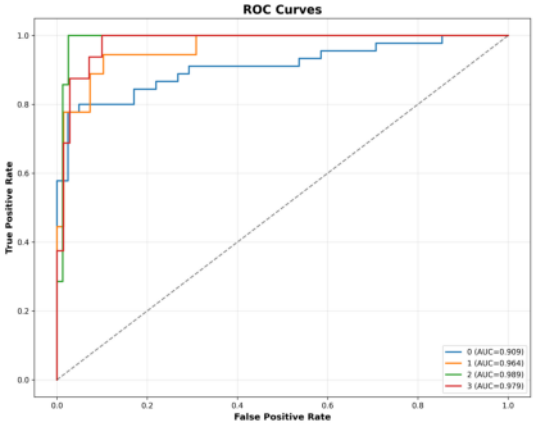
Confusion Matrix



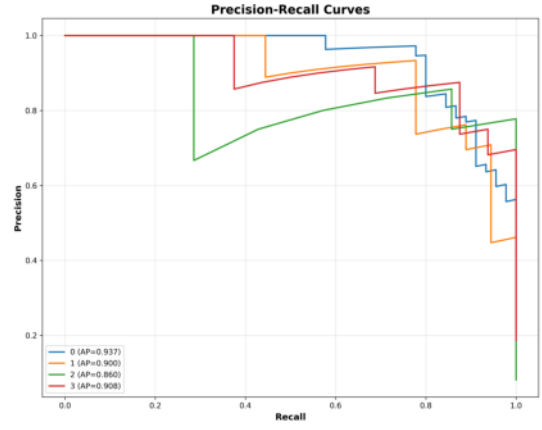
Feature Importance



ROC Curves

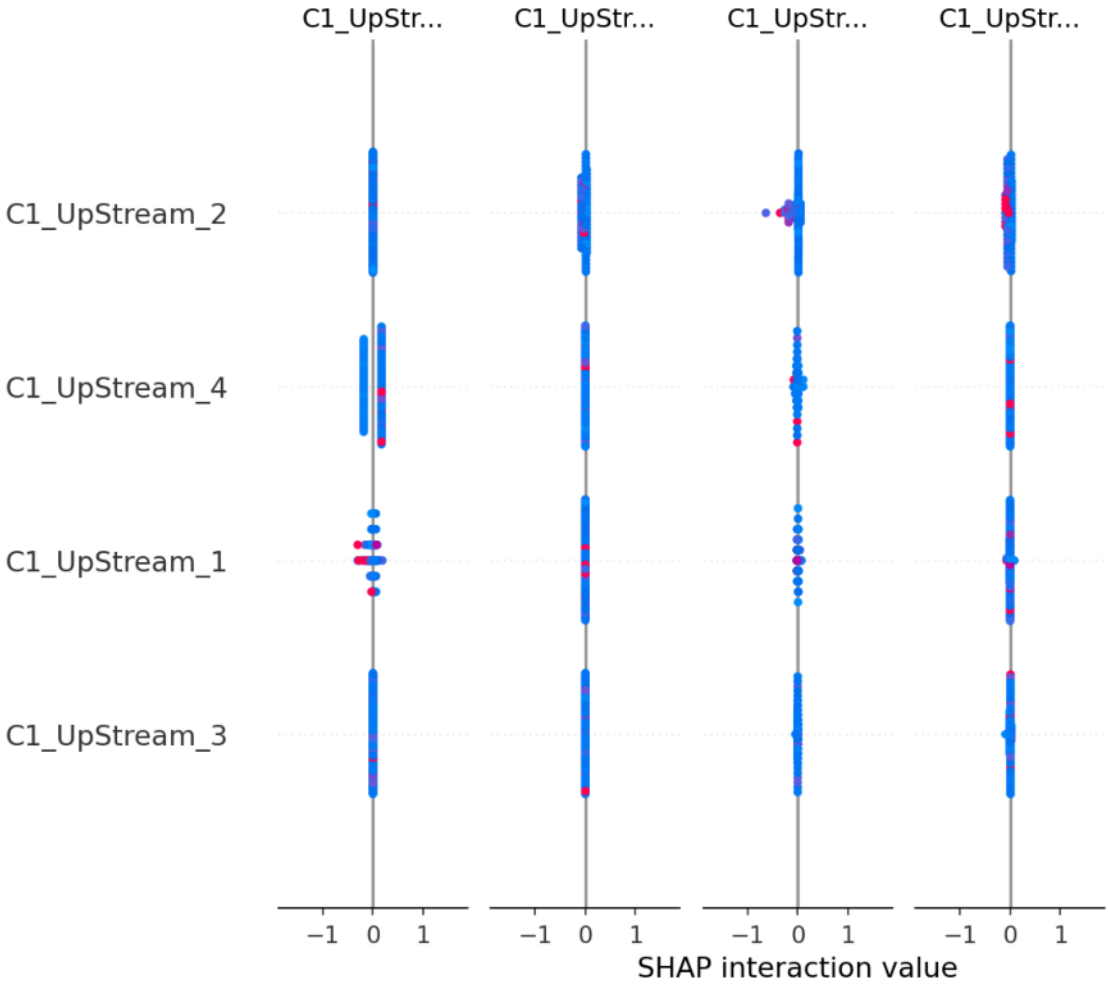


Precision-Recall Curves

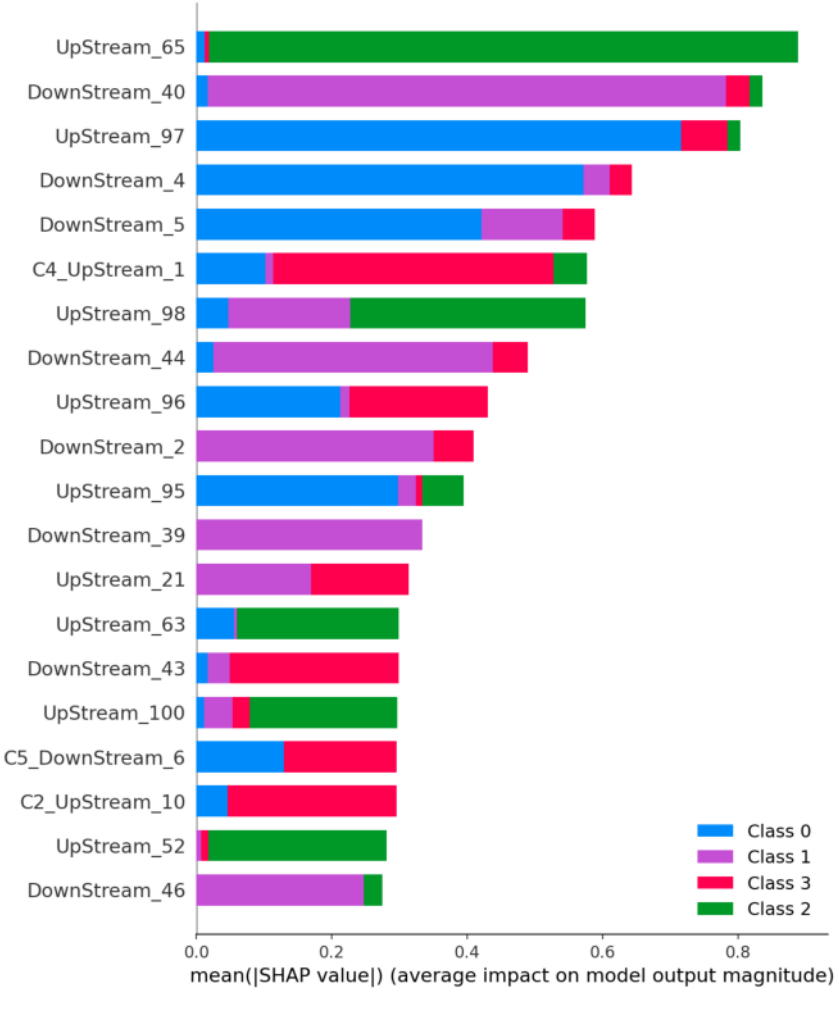


XGBOOST - Original - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

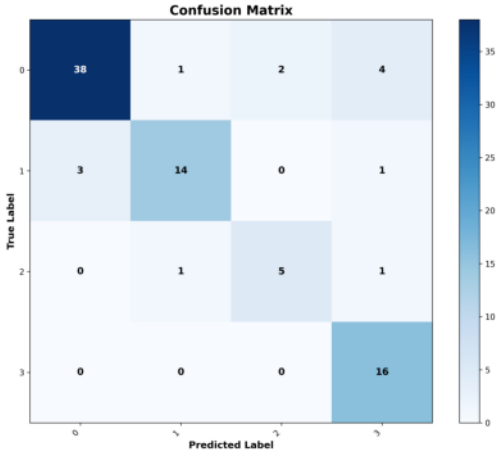


XGBOOST - SMOTE Strategy

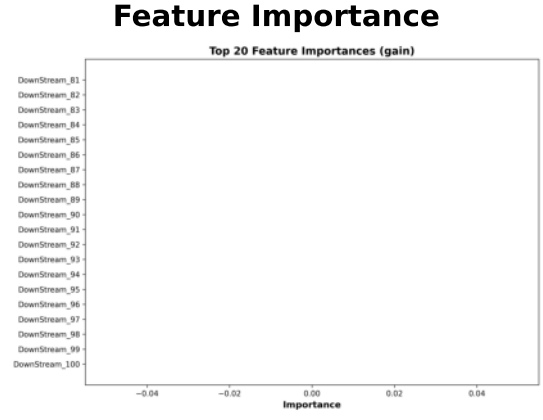
Performance Metrics

Accuracy	0.8488
Balanced Accuracy	0.8341
F1-Score (Macro)	0.8159
MSE	0.6279
MAE	0.2791
Log Loss	0.4737

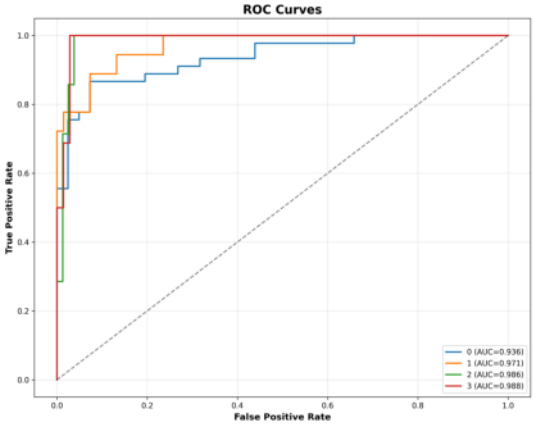
Confusion Matrix



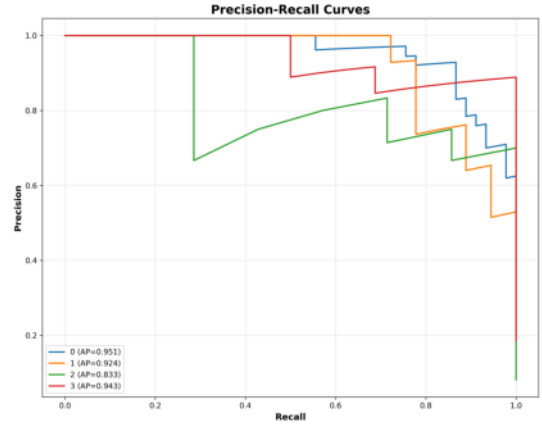
Feature Importance



ROC Curves

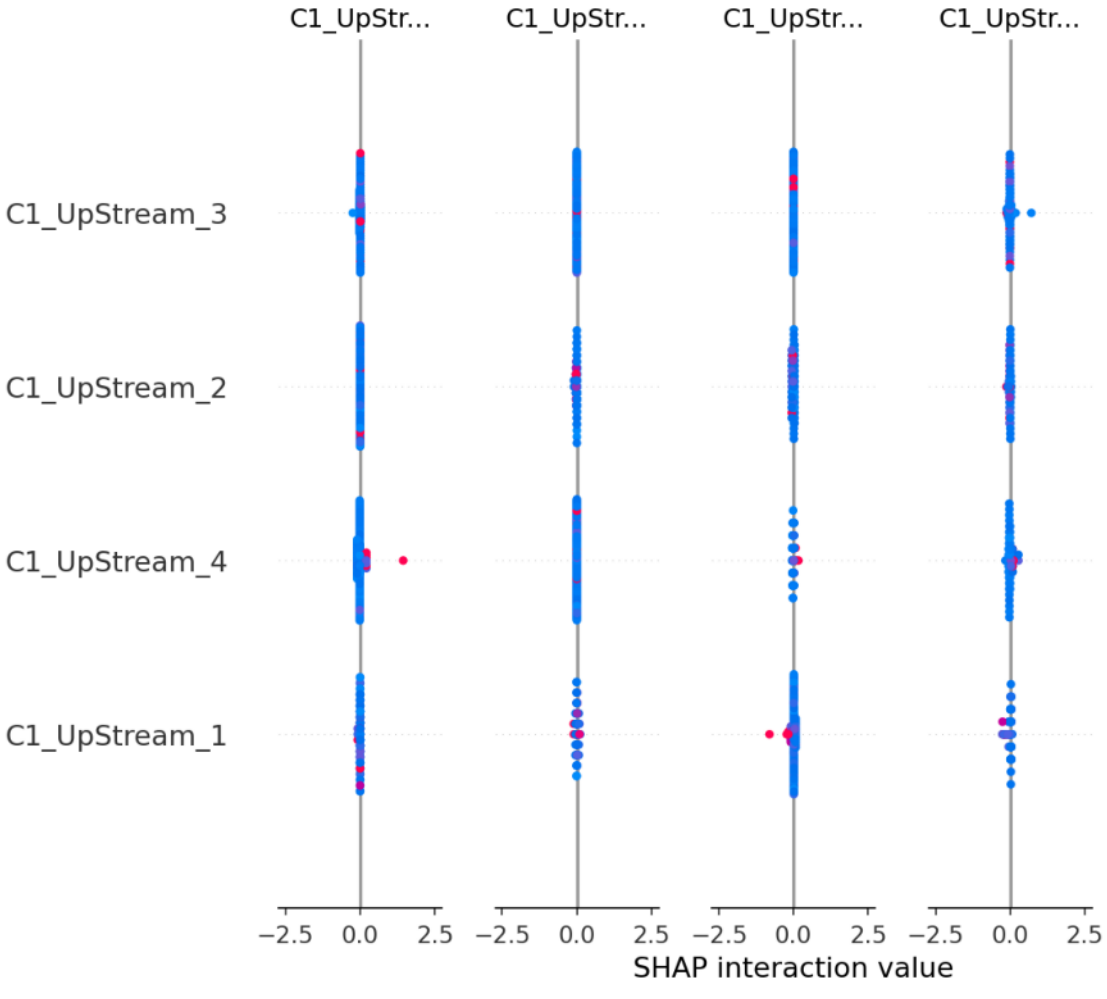


Precision-Recall Curves

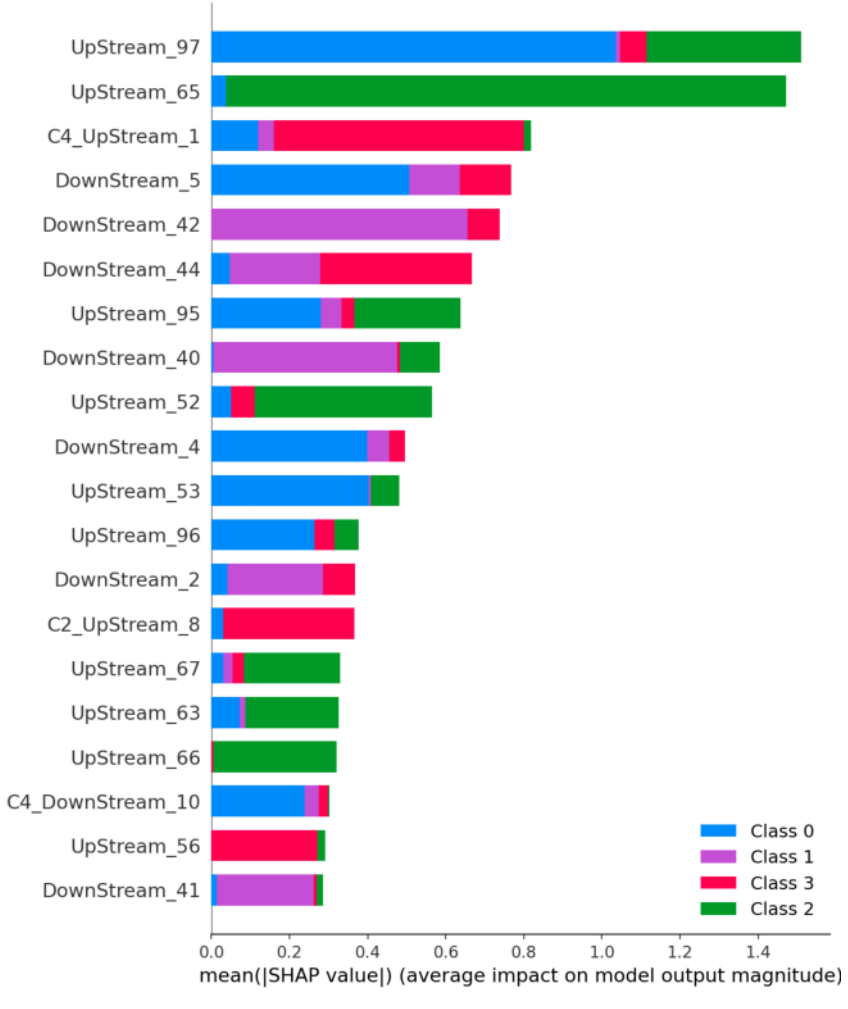


XGBOOST - SMOTE - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

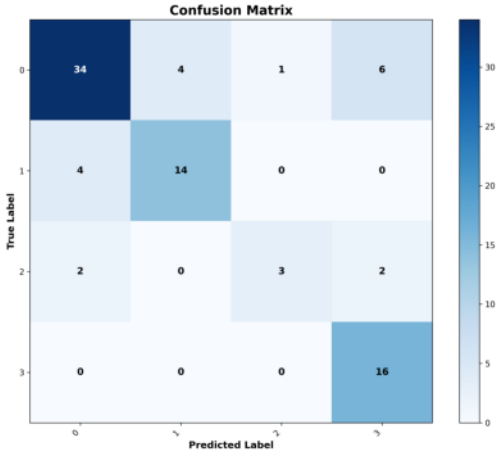


XGBOOST - RandomOverSampler Strategy

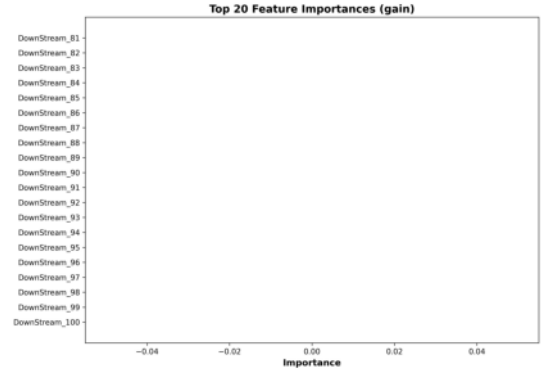
Performance Metrics

Accuracy	0.7791
Balanced Accuracy	0.7405
F1-Score (Macro)	0.7308
MSE	0.8837
MAE	0.3953
Log Loss	0.5527

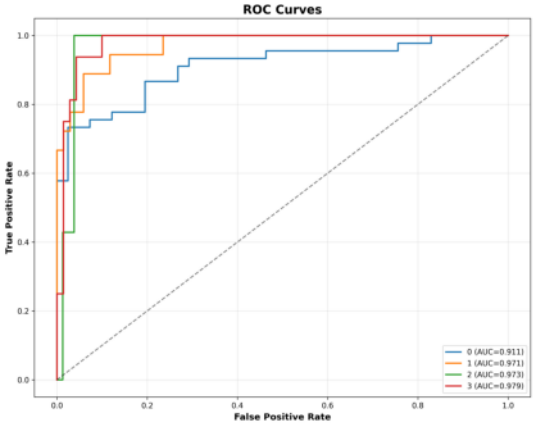
Confusion Matrix



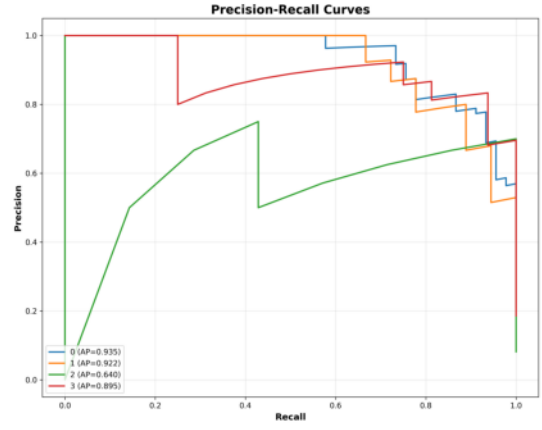
Feature Importance



ROC Curves

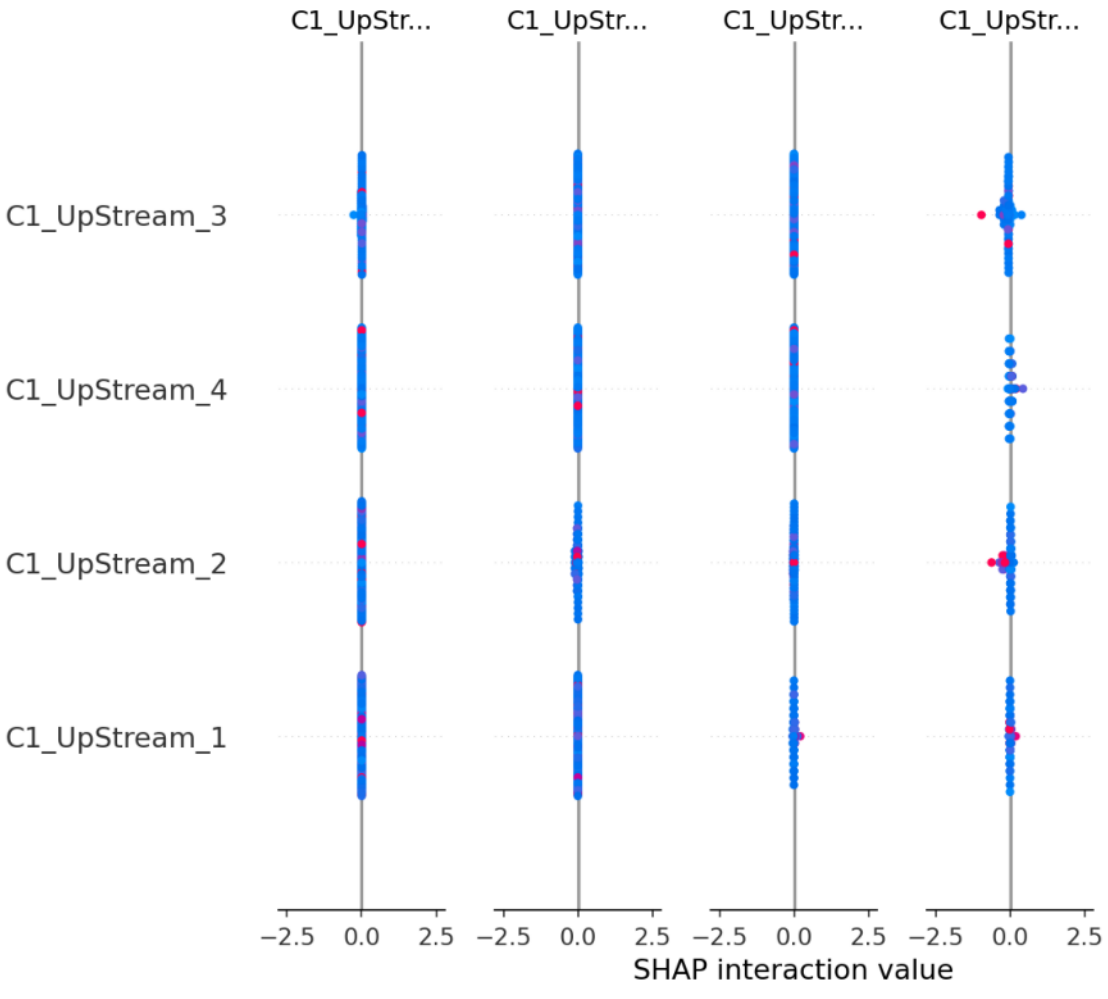


Precision-Recall Curves

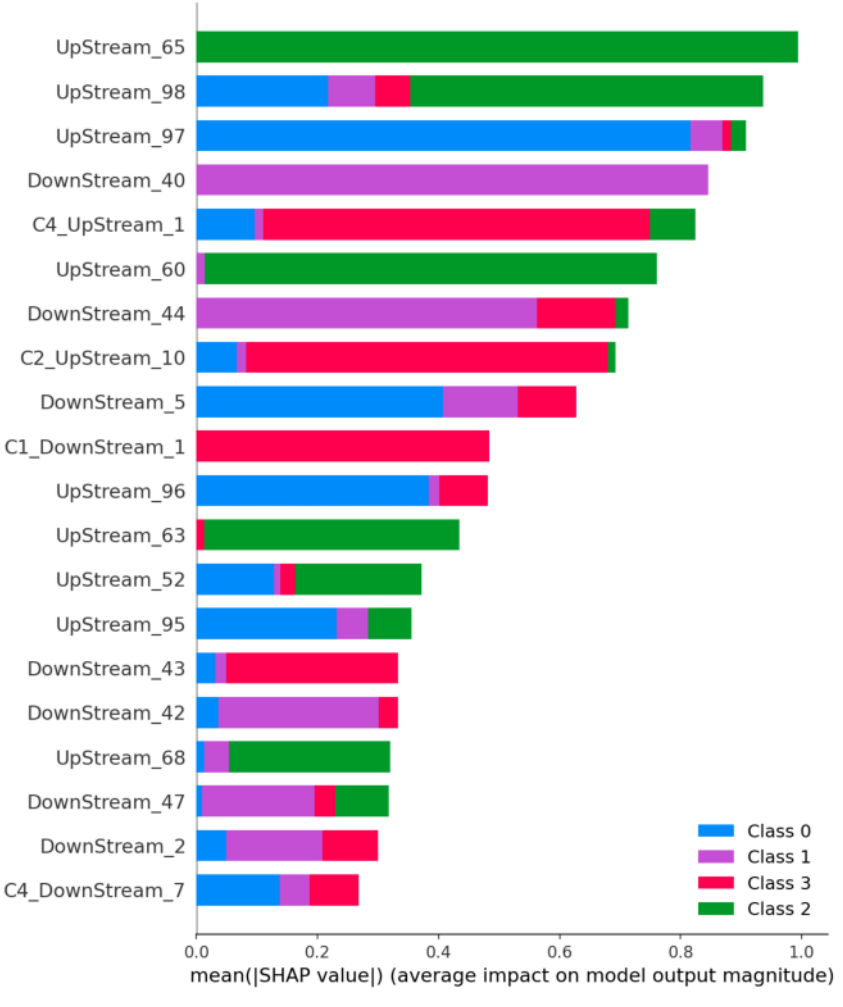


XGBOOST - RandomOverSampler - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

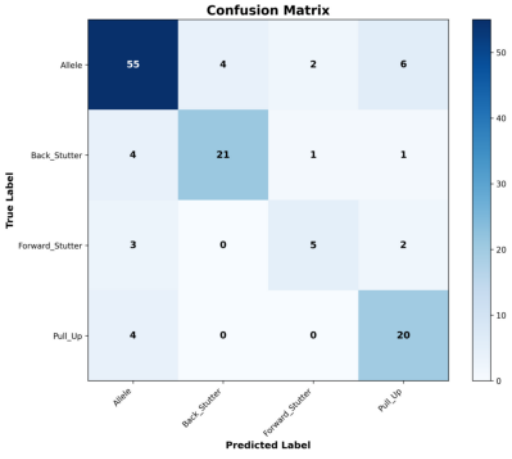


RANDOMFOREST - No Resampling Strategy

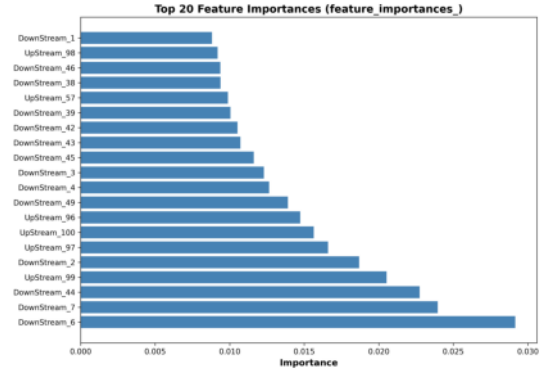
Performance Metrics

Accuracy	0.7891
Balanced Accuracy	0.7330
F1-Score (Macro)	0.7363
MSE	0.9766
MAE	0.4141
Log Loss	0.5895

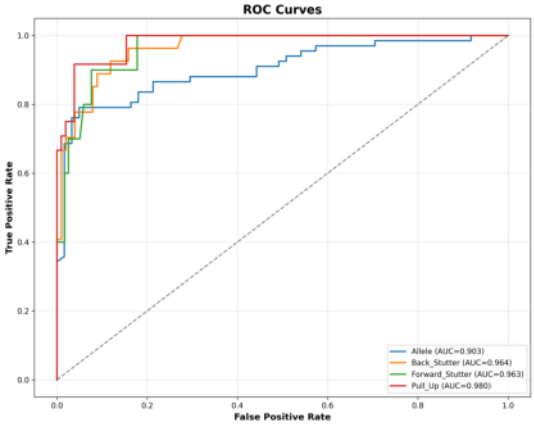
Confusion Matrix



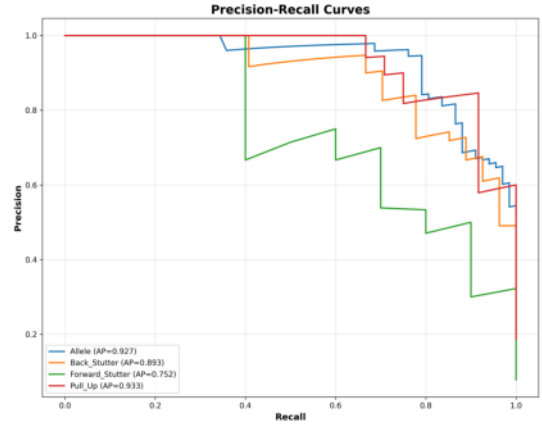
Feature Importance



ROC Curves

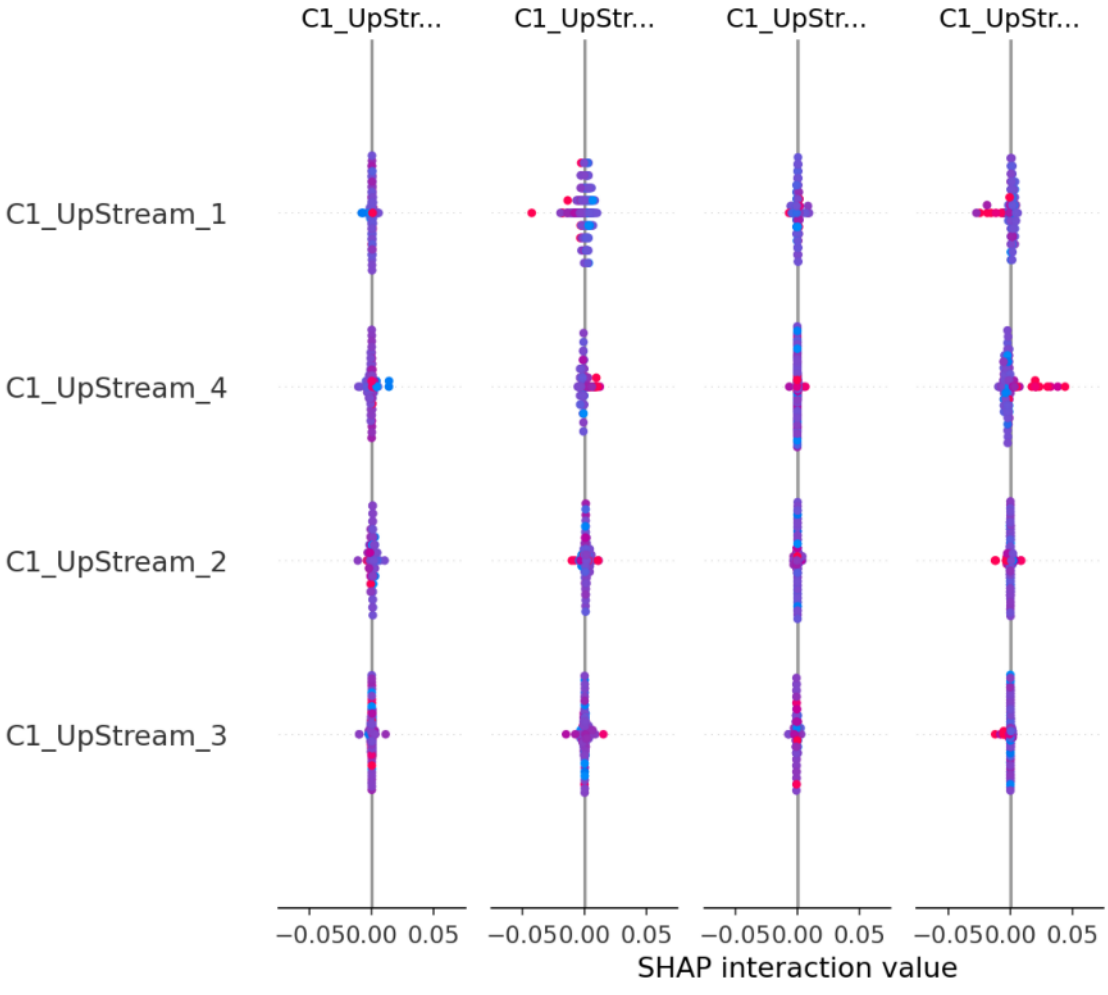


Precision-Recall Curves

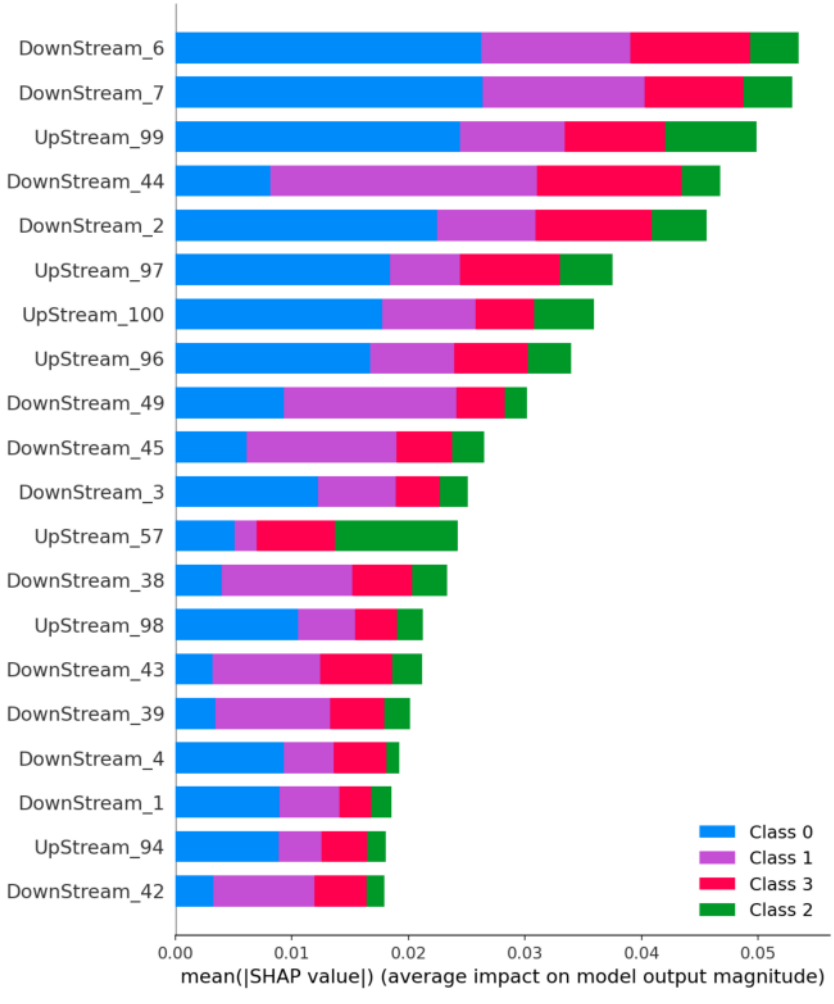


RANDOMFOREST - No Resampling - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

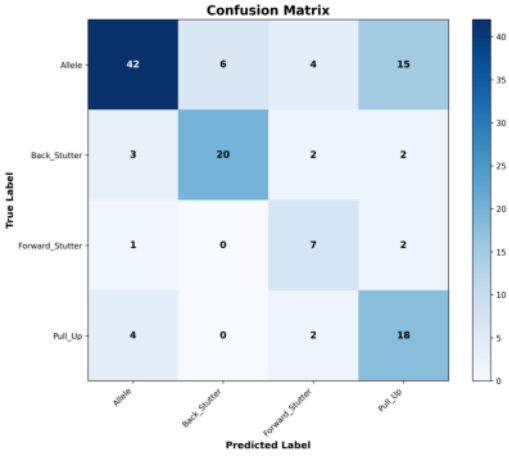


RANDOMFOREST - Class Weights Strategy

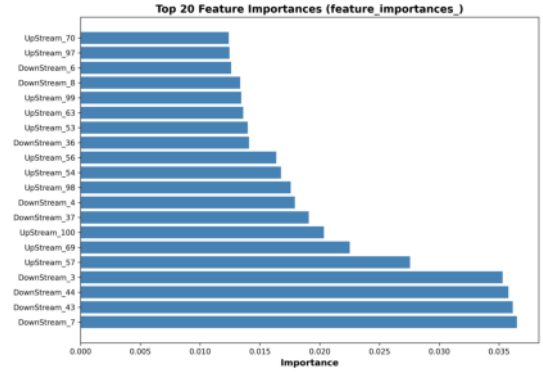
Performance Metrics

Accuracy	0.6797
Balanced Accuracy	0.7044
F1-Score (Macro)	0.6557
MSE	1.6719
MAE	0.6719
Log Loss	0.7949

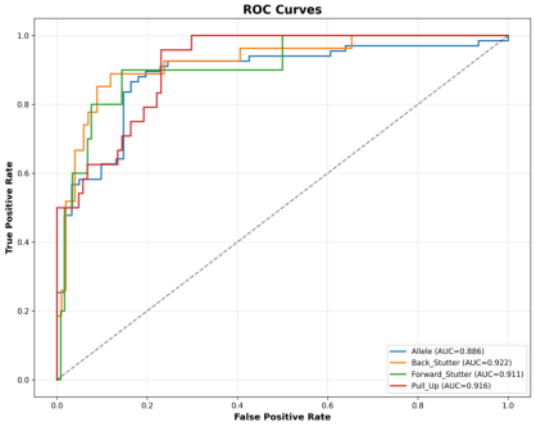
Confusion Matrix



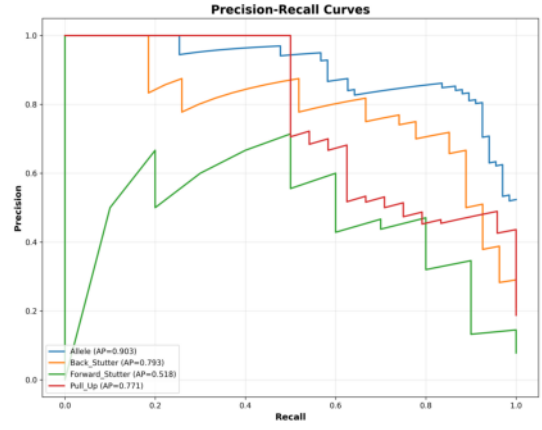
Feature Importance



ROC Curves

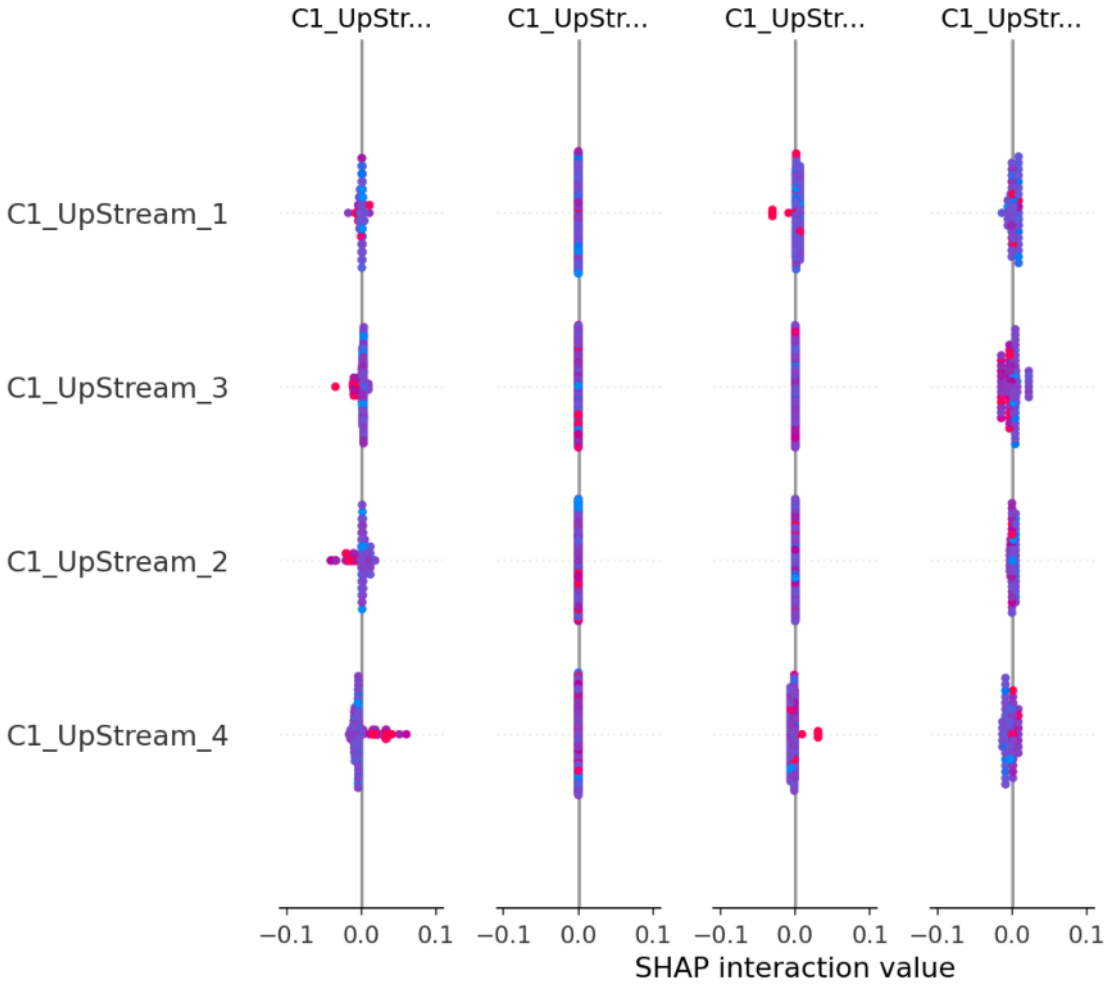


Precision-Recall Curves

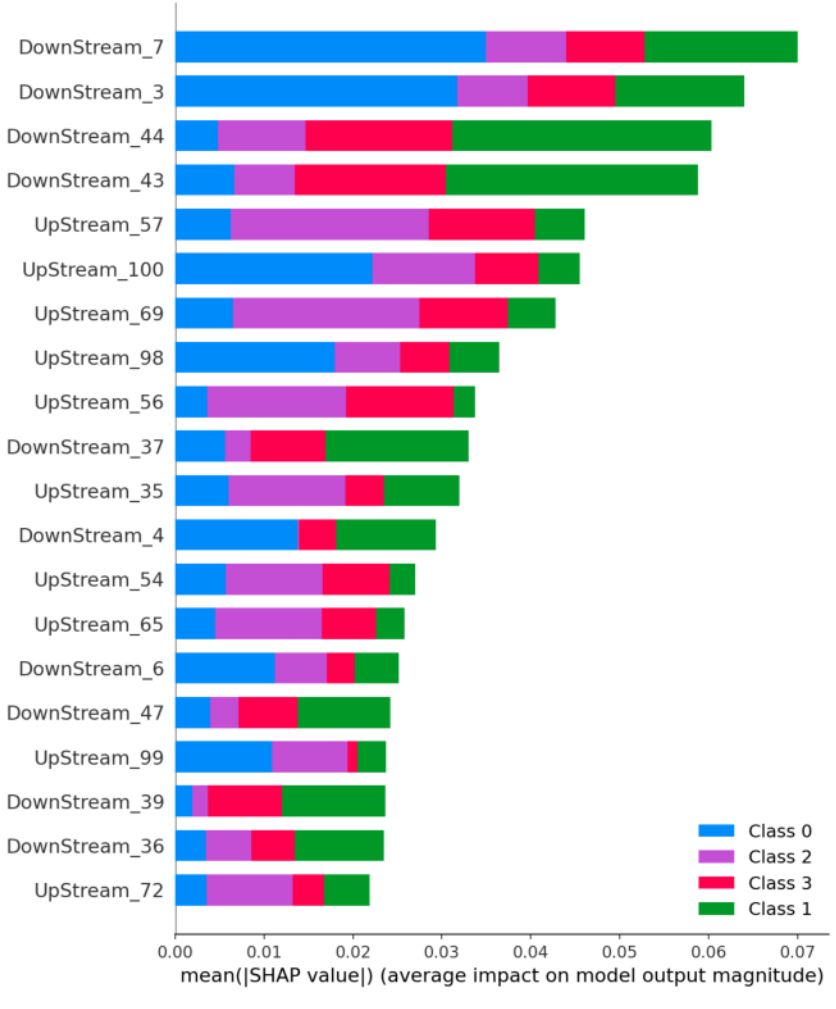


RANDOMFOREST - Class Weights - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

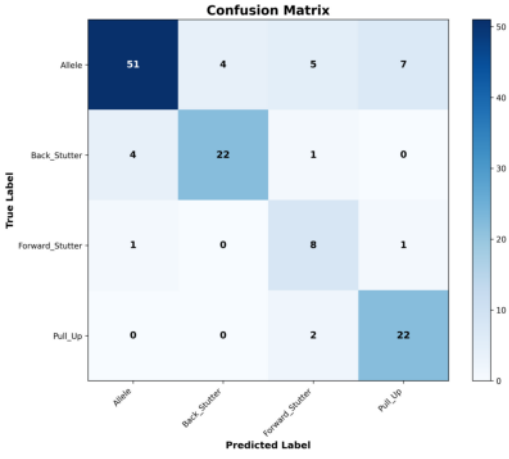


RANDOMFOREST - SMOTE Strategy

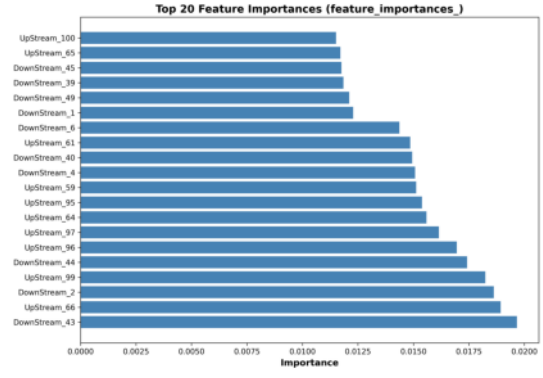
Performance Metrics

Accuracy	0.8047
Balanced Accuracy	0.8232
F1-Score (Macro)	0.7724
MSE	0.7734
MAE	0.3516
Log Loss	0.5574

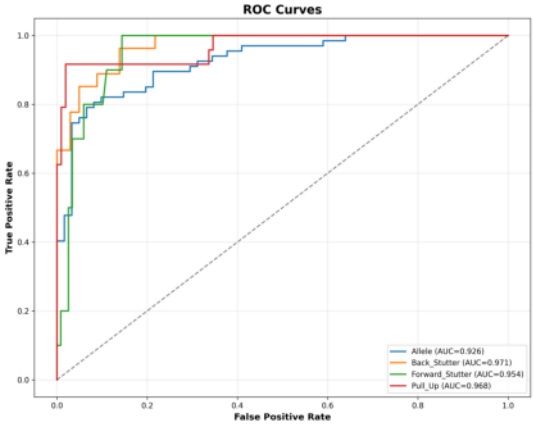
Confusion Matrix



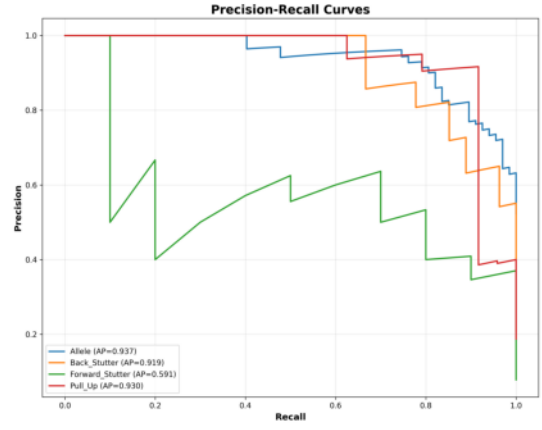
Feature Importance



ROC Curves

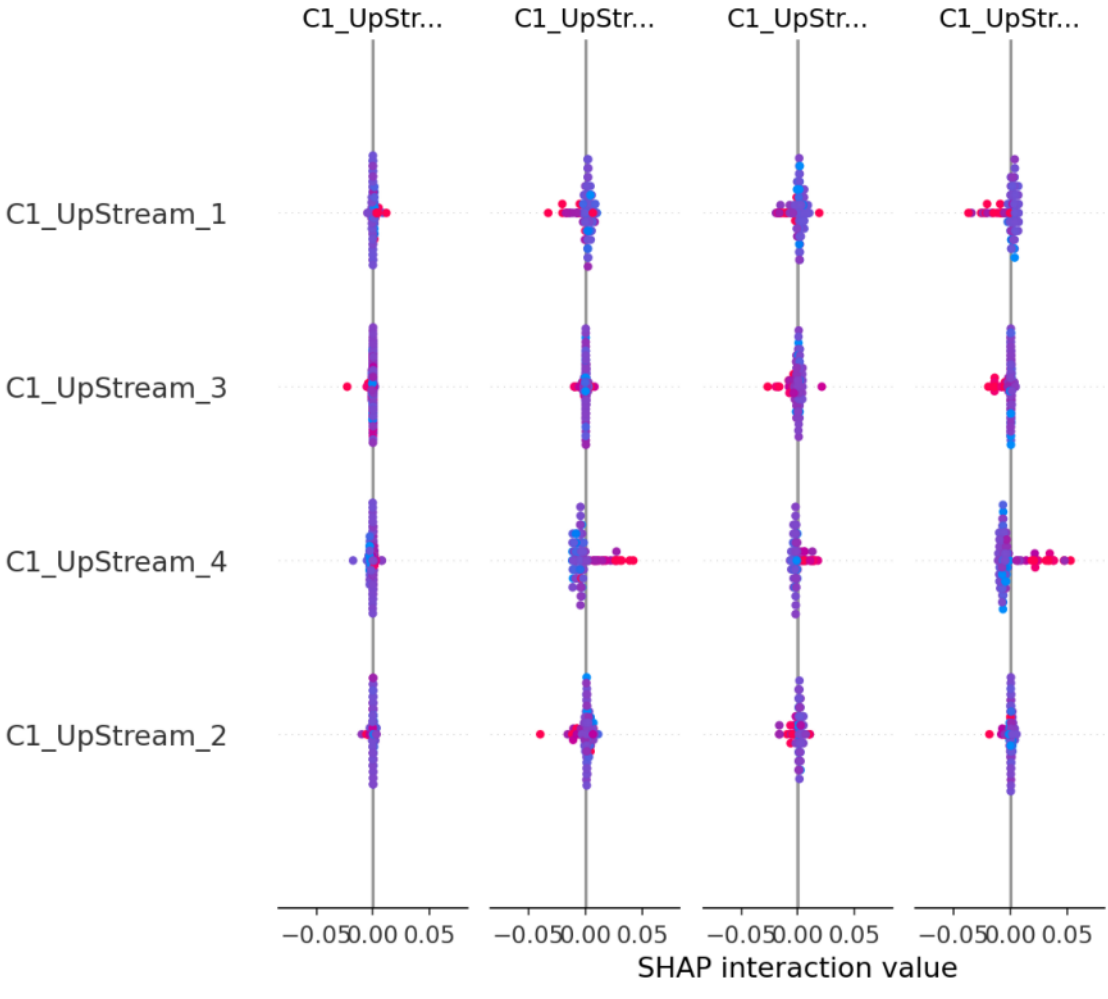


Precision-Recall Curves

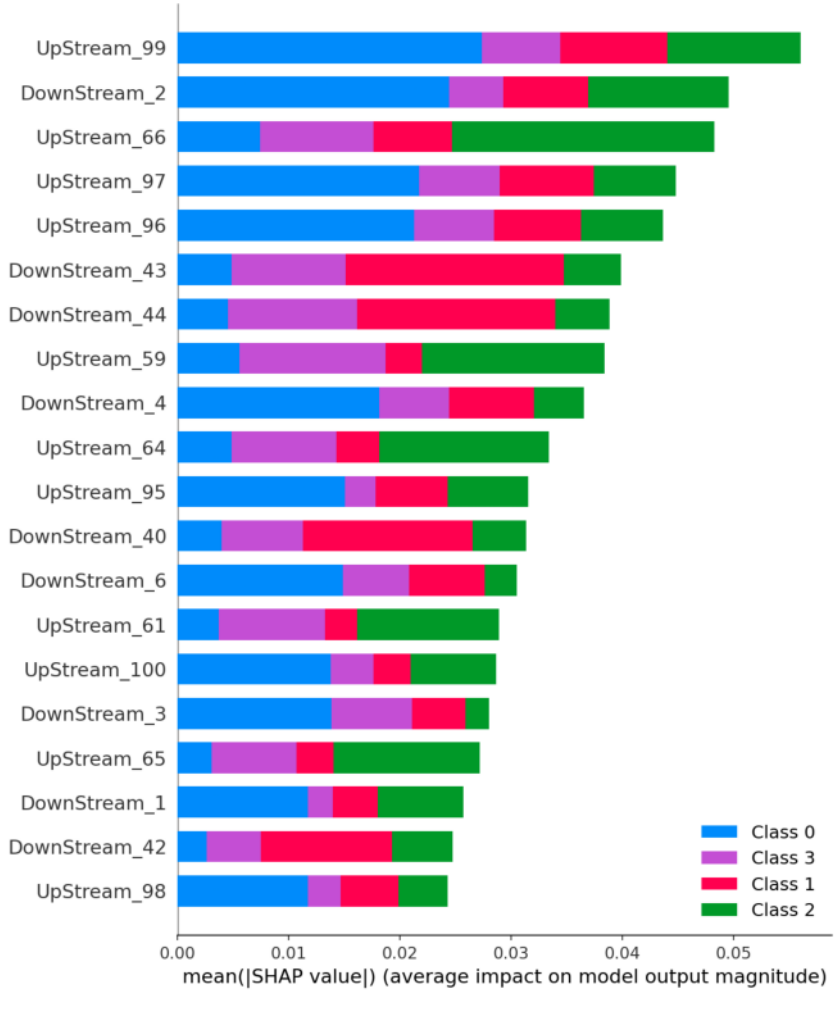


RANDOMFOREST - SMOTE - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

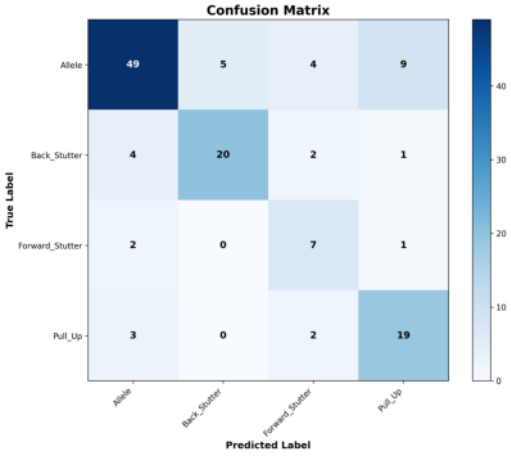


RANDOMFOREST - Borderline SMOTE Strategy

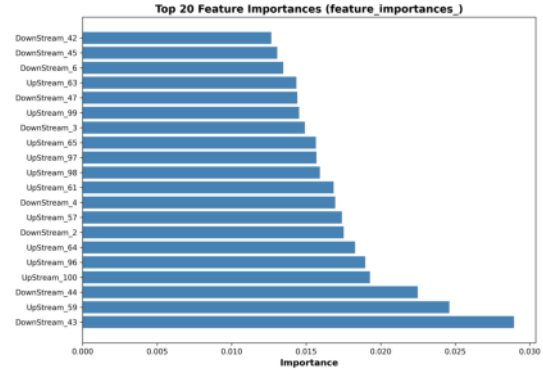
Performance Metrics

Accuracy	0.7422
Balanced Accuracy	0.7409
F1-Score (Macro)	0.7042
MSE	1.1719
MAE	0.5000
Log Loss	0.5580

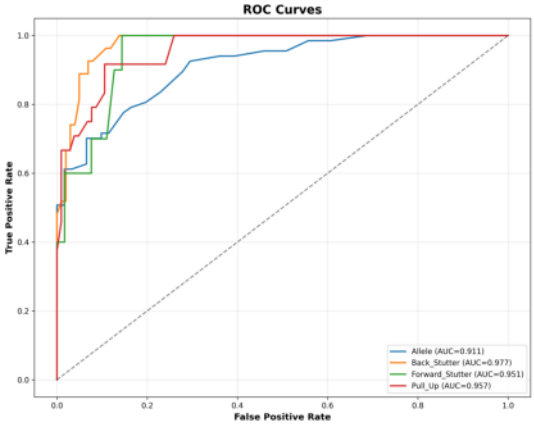
Confusion Matrix



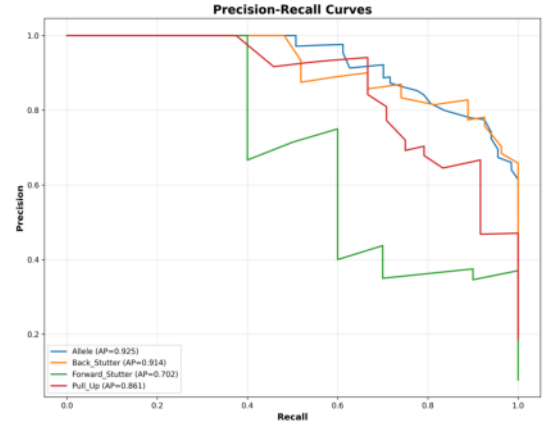
Feature Importance



ROC Curves

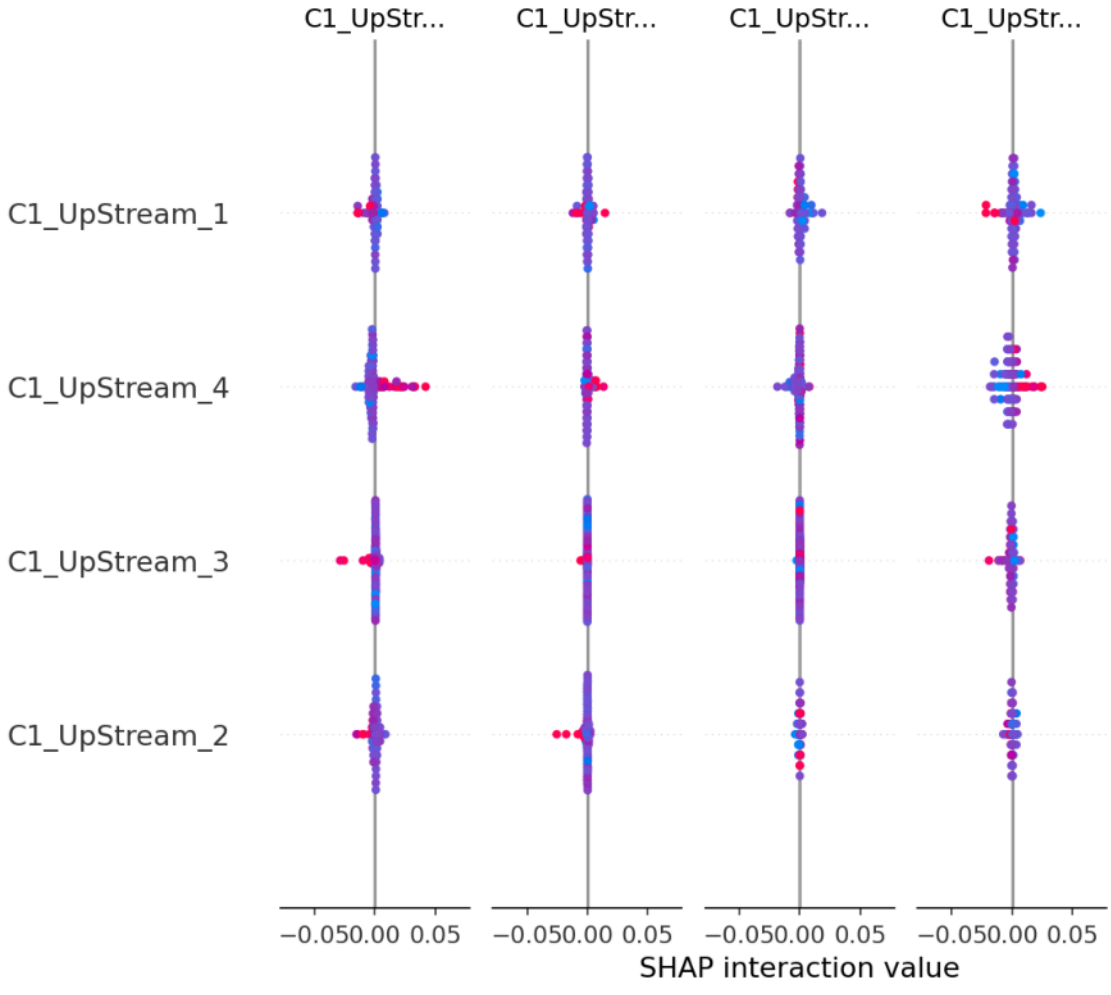


Precision-Recall Curves

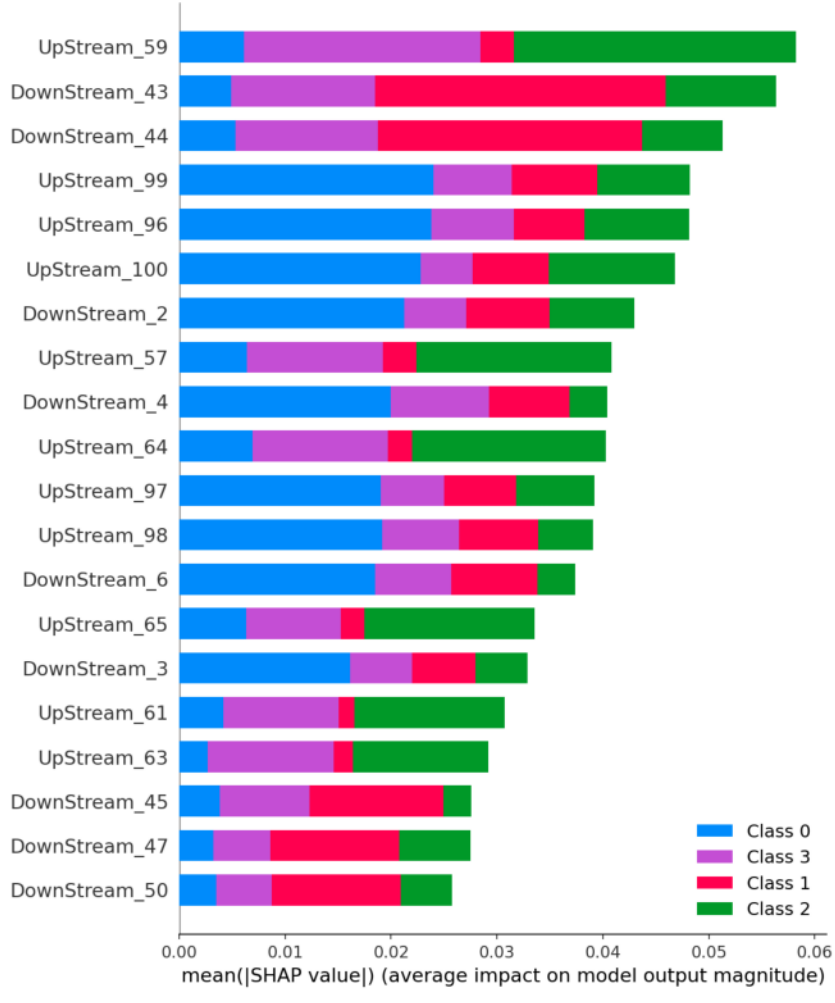


RANDOMFOREST - Borderline SMOTE - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

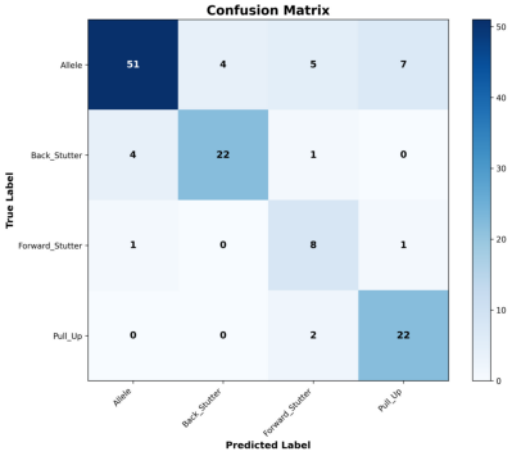


RANDOMFOREST - SVM SMOTE Strategy

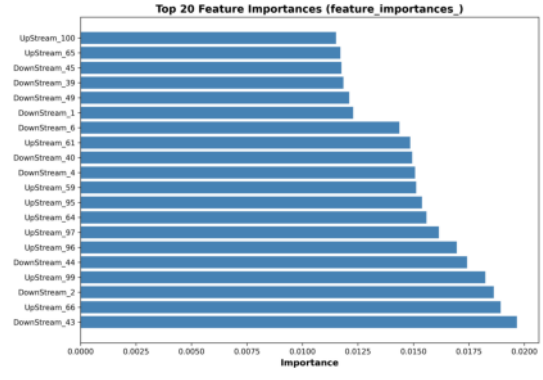
Performance Metrics

Accuracy	0.7812
Balanced Accuracy	0.8120
F1-Score (Macro)	0.7641
MSE	1.0859
MAE	0.4453
Log Loss	0.5641

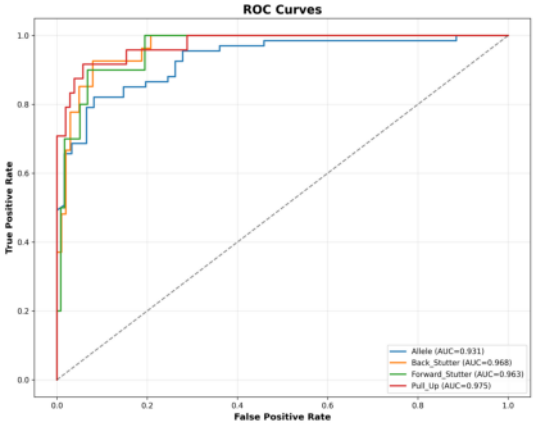
Confusion Matrix



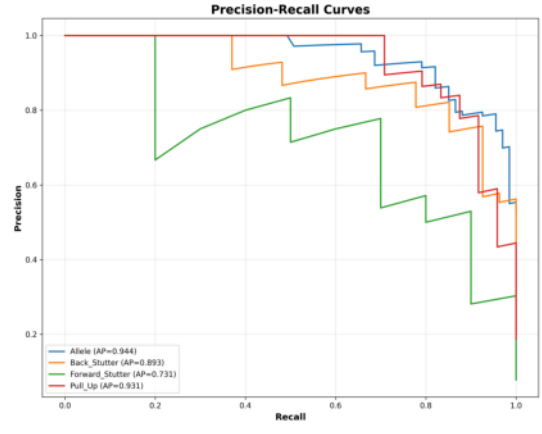
Feature Importance



ROC Curves

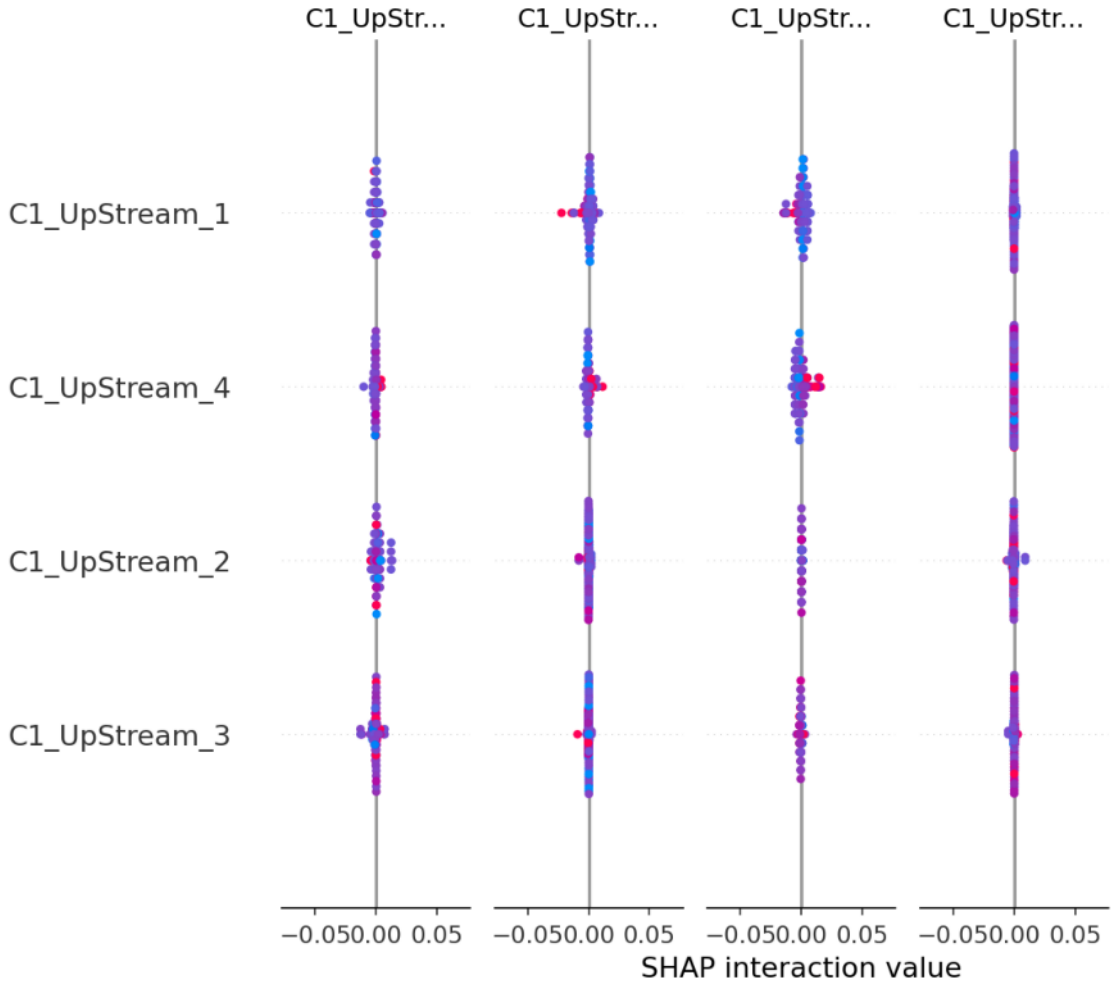


Precision-Recall Curves

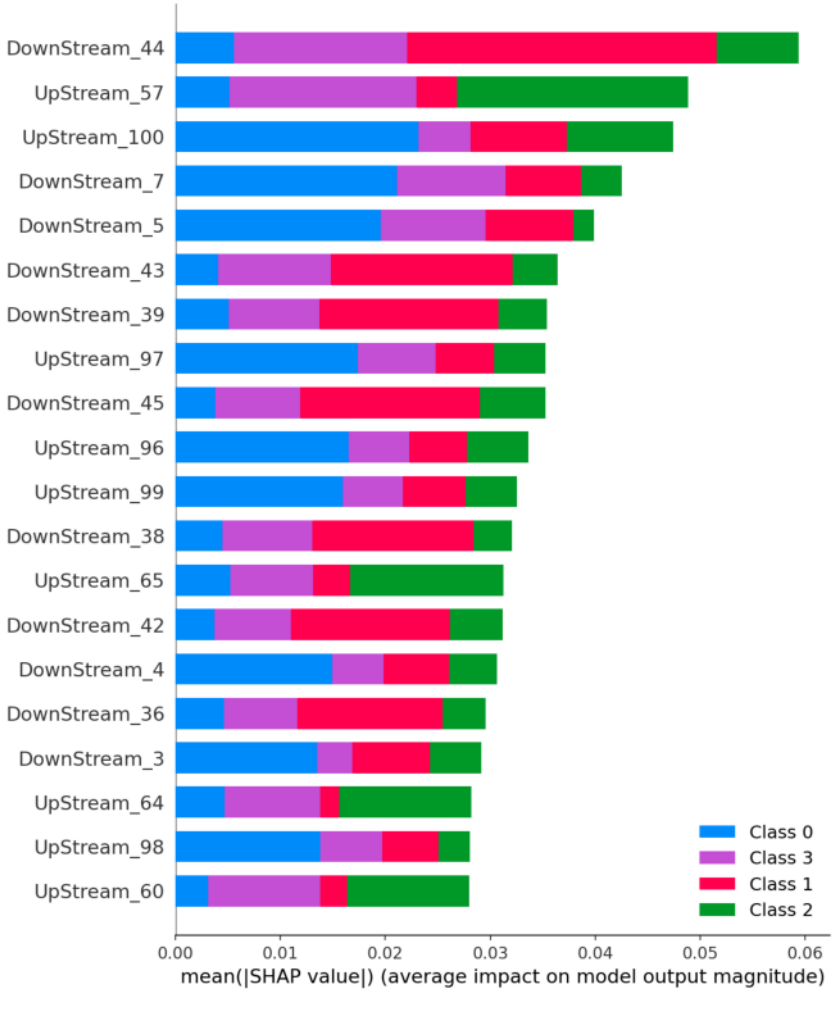


RANDOMFOREST - SVM SMOTE - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

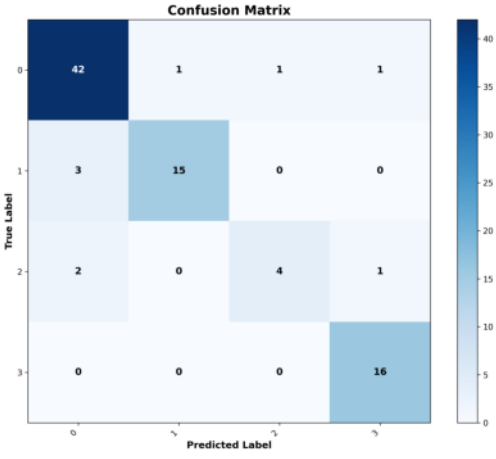


TABPFN - Original Strategy

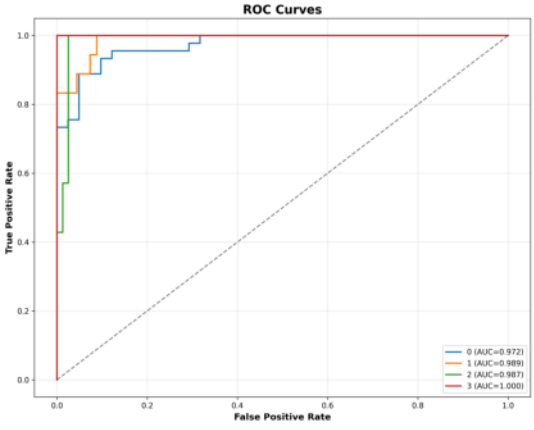
Performance Metrics

Accuracy	0.8953
Balanced Accuracy	0.8345
F1-Score (Macro)	0.8508
MSE	0.3023
MAE	0.1628
Log Loss	0.2779

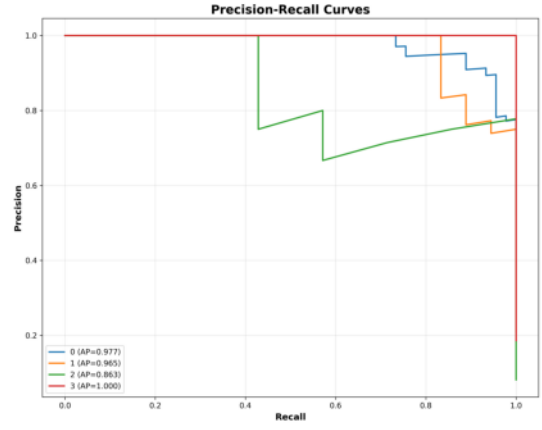
Confusion Matrix



ROC Curves



Precision-Recall Curves

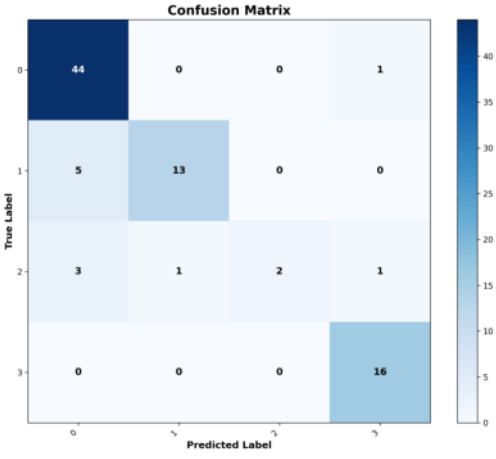


TABPFN - SMOTE Strategy

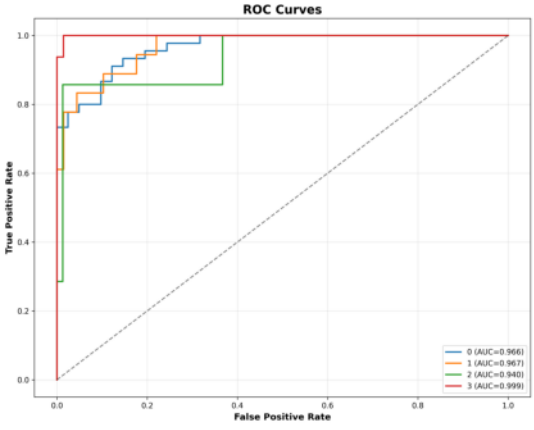
Performance Metrics

Accuracy	0.8721
Balanced Accuracy	0.7464
F1-Score (Macro)	0.7763
MSE	0.3256
MAE	0.1860
Log Loss	0.3843

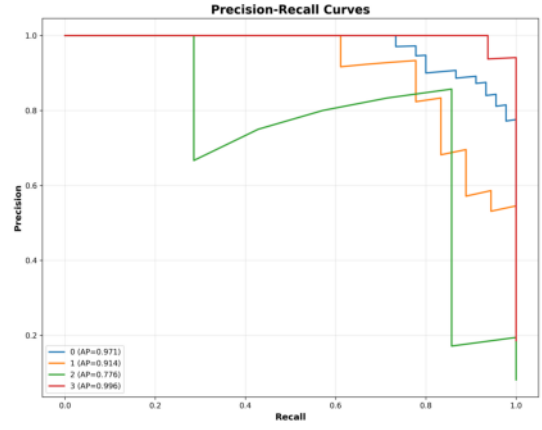
Confusion Matrix



ROC Curves



Precision-Recall Curves

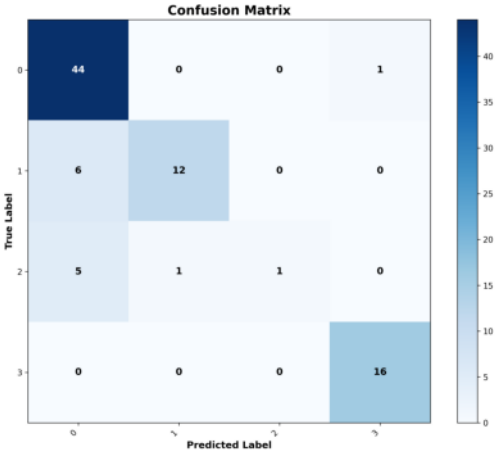


TABPFN - RandomOverSampler Strategy

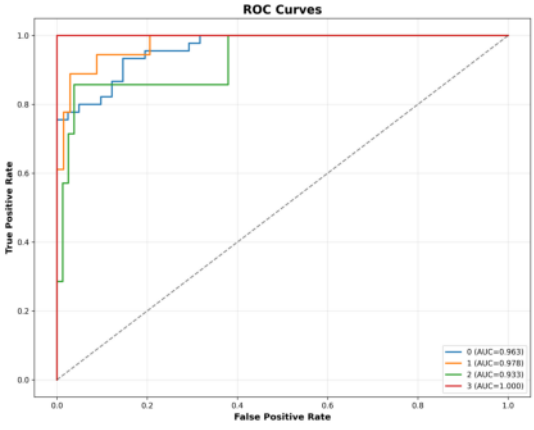
Performance Metrics

Accuracy	0.8488
Balanced Accuracy	0.6968
F1-Score (Macro)	0.7185
MSE	0.4186
MAE	0.2326
Log Loss	0.4285

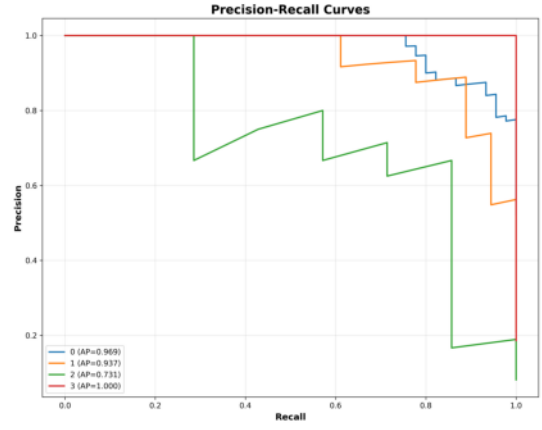
Confusion Matrix



ROC Curves



Precision-Recall Curves

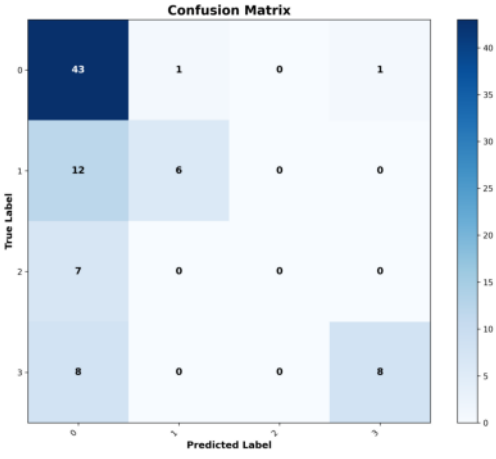


SVM - Original Strategy

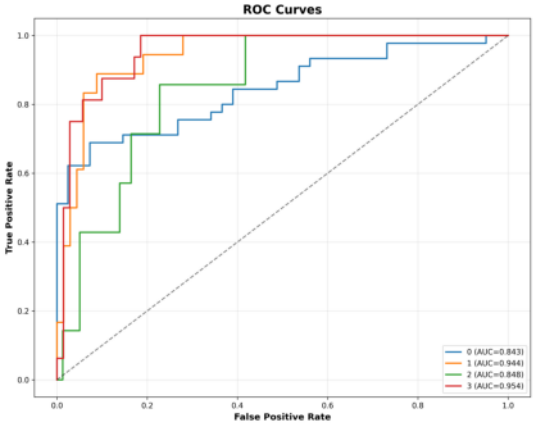
Performance Metrics

Accuracy	0.6628
Balanced Accuracy	0.4472
F1-Score (Macro)	0.4670
MSE	1.4186
MAE	0.6279
Log Loss	0.7420

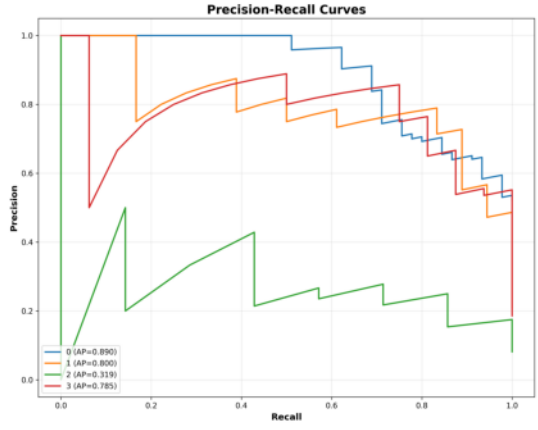
Confusion Matrix



ROC Curves



Precision-Recall Curves

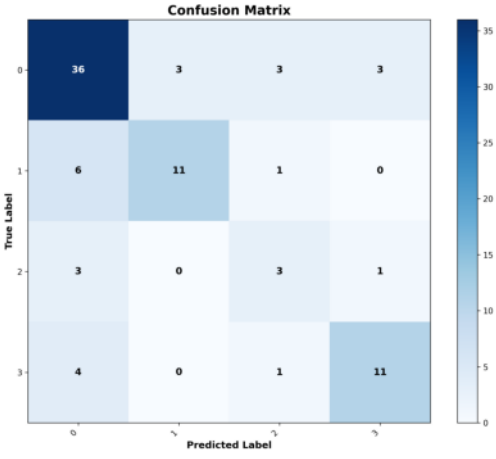


SVM - SMOTE Strategy

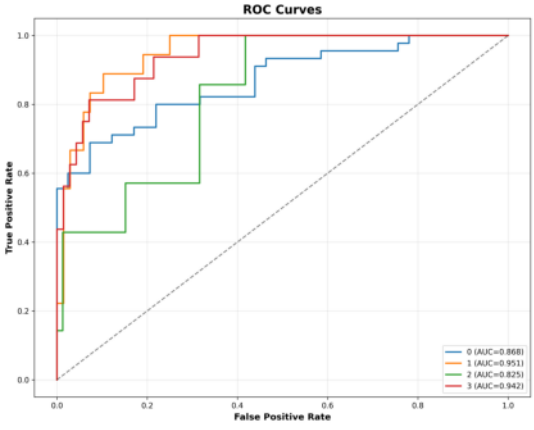
Performance Metrics

Accuracy	0.7093
Balanced Accuracy	0.6318
F1-Score (Macro)	0.6408
MSE	1.1512
MAE	0.5233
Log Loss	0.7460

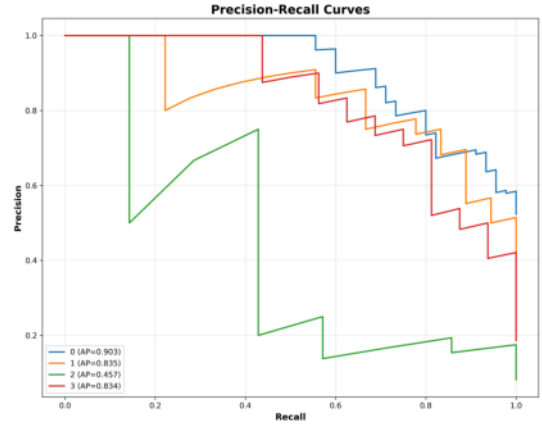
Confusion Matrix



ROC Curves



Precision-Recall Curves

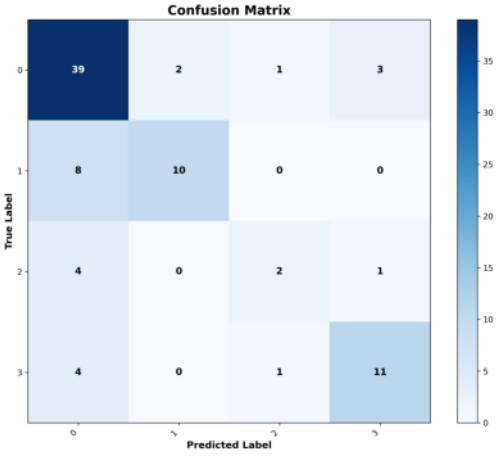


SVM - RandomOverSampler Strategy

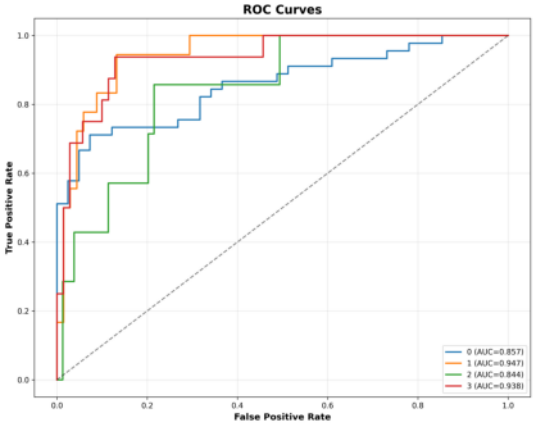
Performance Metrics

Accuracy	0.7209
Balanced Accuracy	0.5989
F1-Score (Macro)	0.6300
MSE	1.1047
MAE	0.5000
Log Loss	0.7540

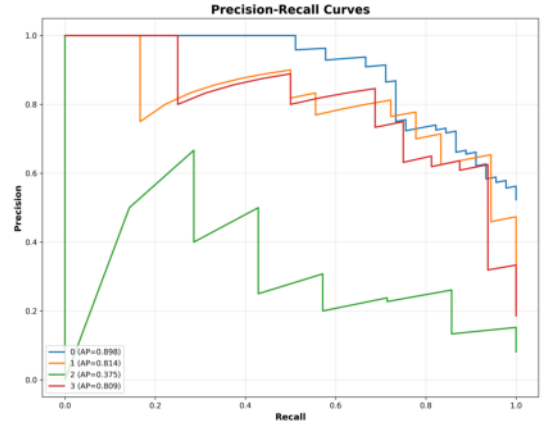
Confusion Matrix



ROC Curves



Precision-Recall Curves

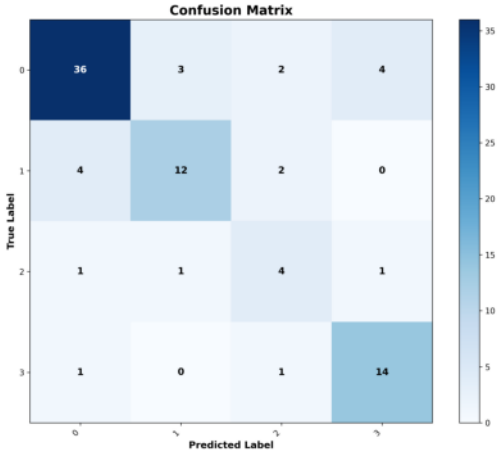


LOGISTIC_REGRESSION - Original Strategy

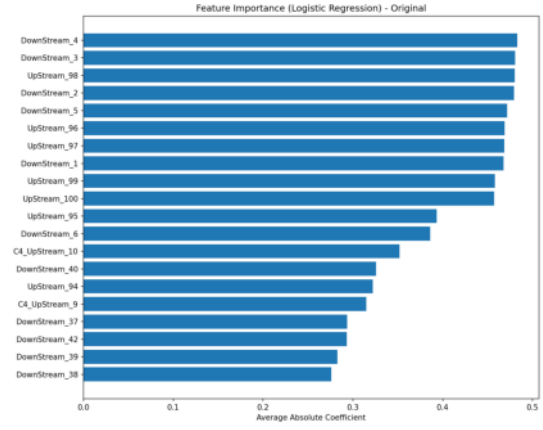
Performance Metrics

Accuracy	0.7674
Balanced Accuracy	0.7283
F1-Score (Macro)	0.7084
MSE	0.8023
MAE	0.3837
Log Loss	1.4218

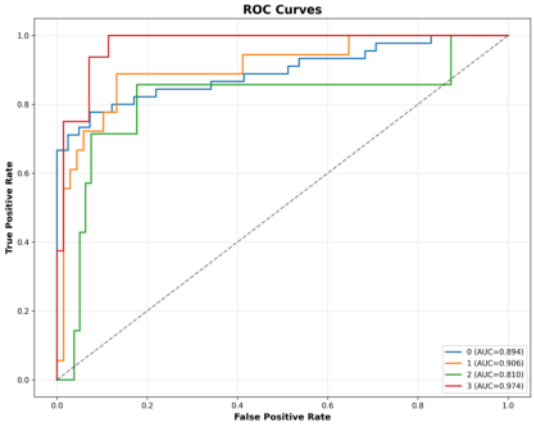
Confusion Matrix



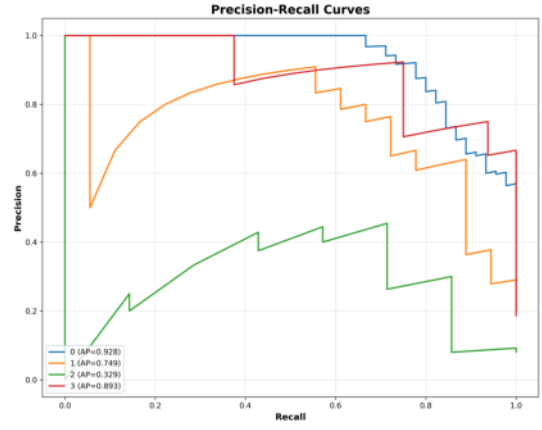
Feature Importance



ROC Curves

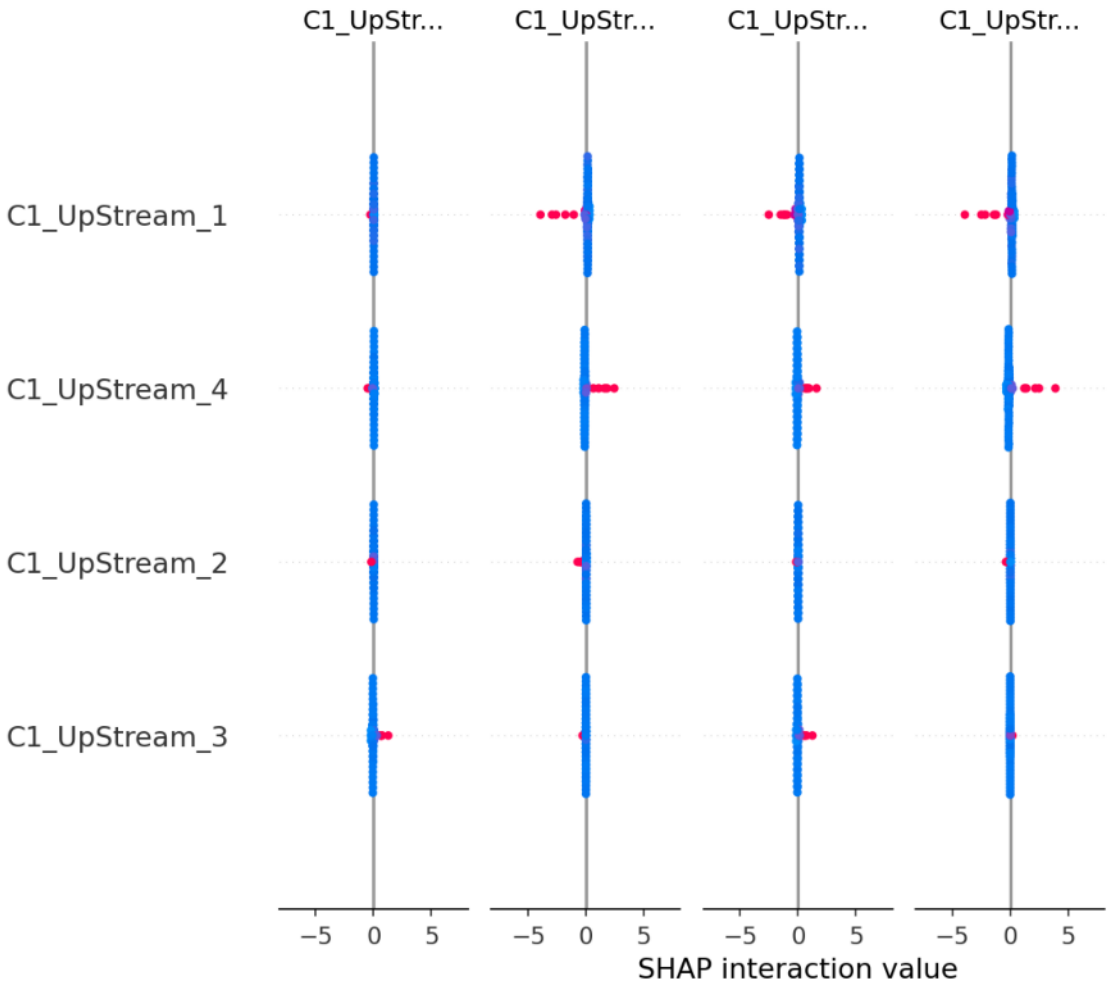


Precision-Recall Curves

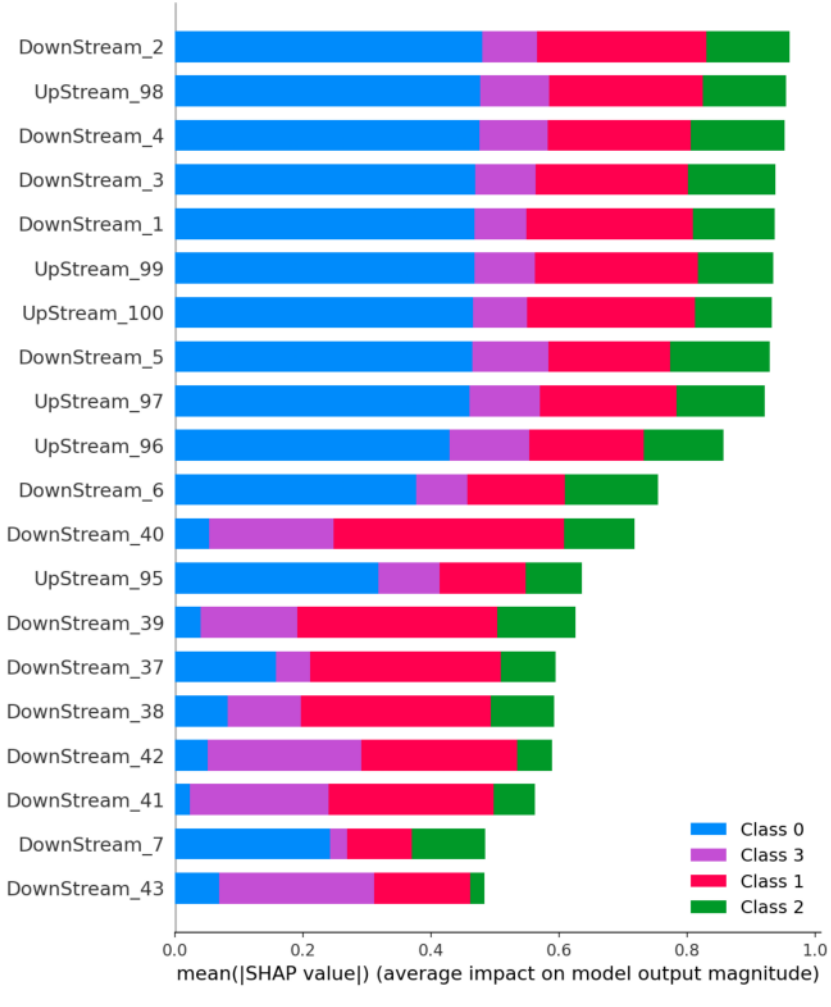


LOGISTIC_REGRESSION - Original - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

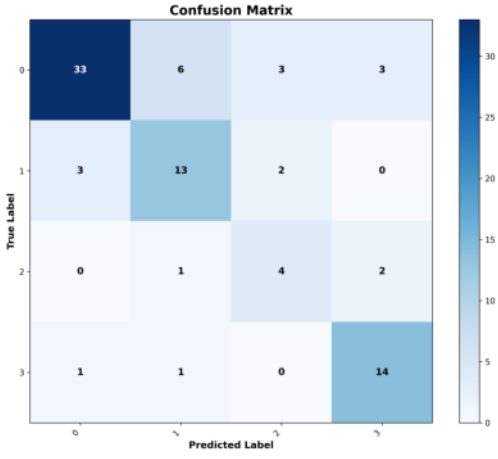


LOGISTIC_REGRESSION - SMOTE Strategy

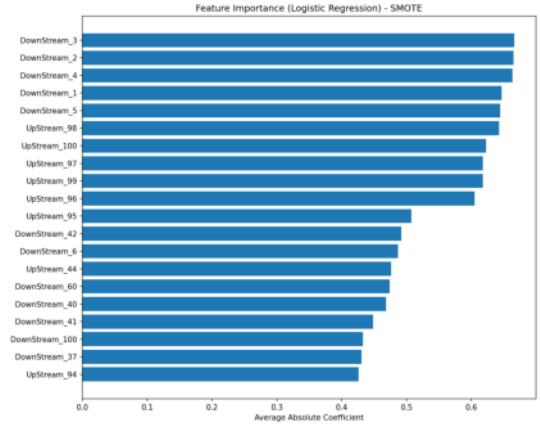
Performance Metrics

Accuracy	0.7442
Balanced Accuracy	0.7255
F1-Score (Macro)	0.6929
MSE	0.7674
MAE	0.3953
Log Loss	2.3497

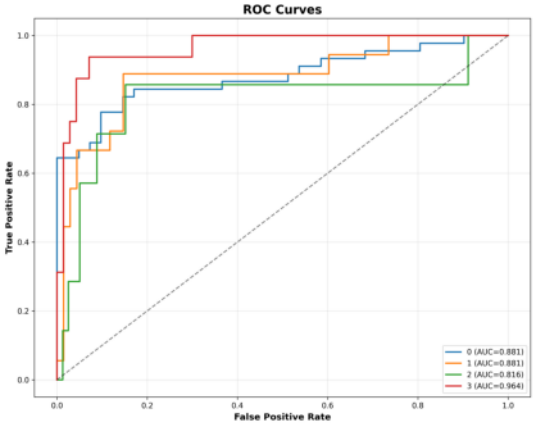
Confusion Matrix



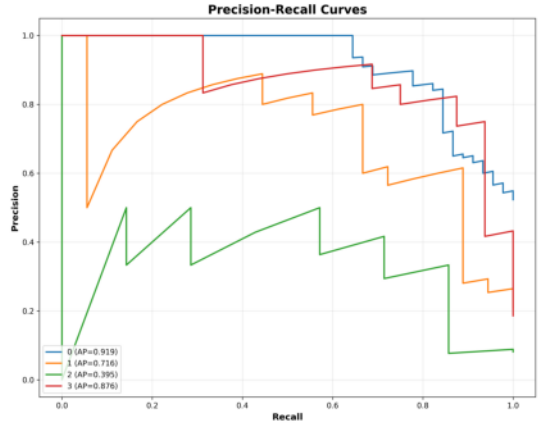
Feature Importance



ROC Curves

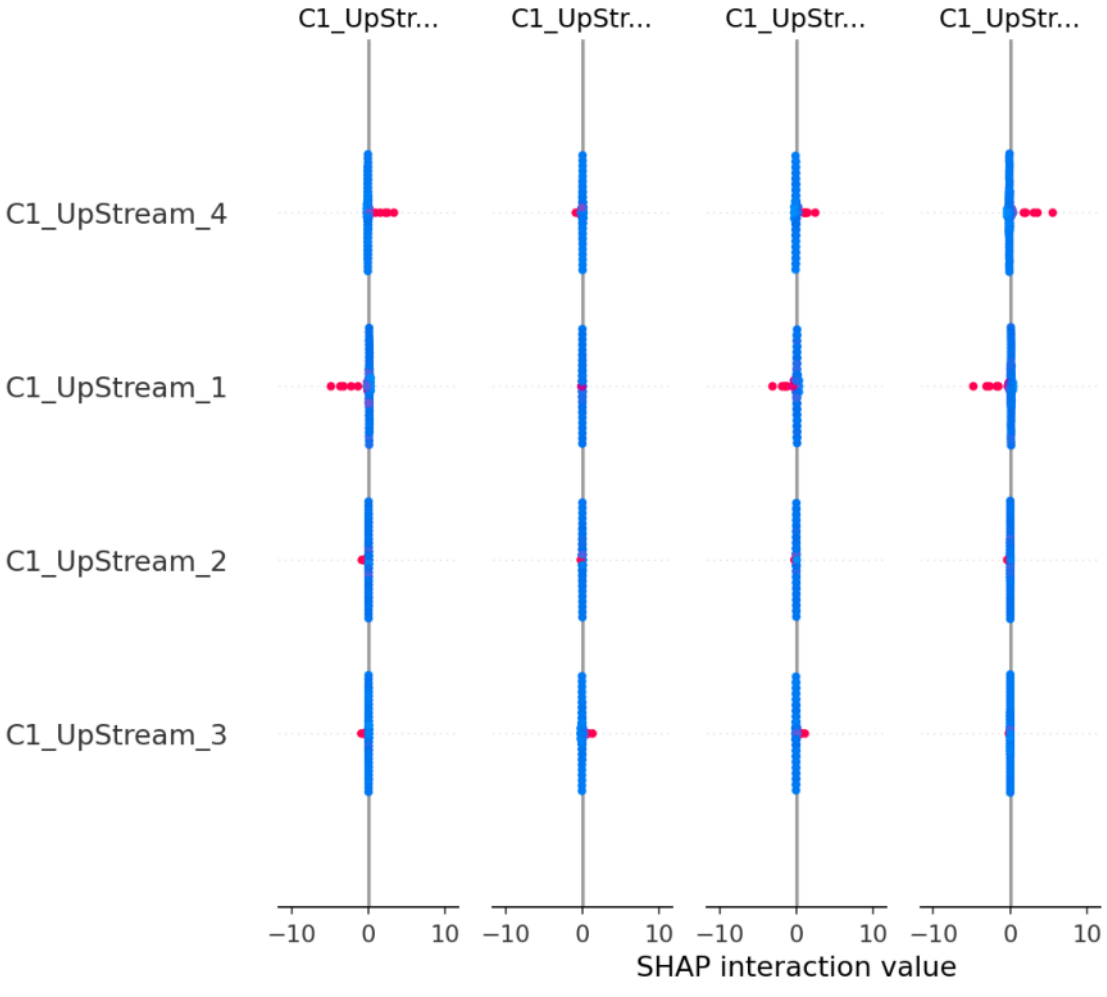


Precision-Recall Curves

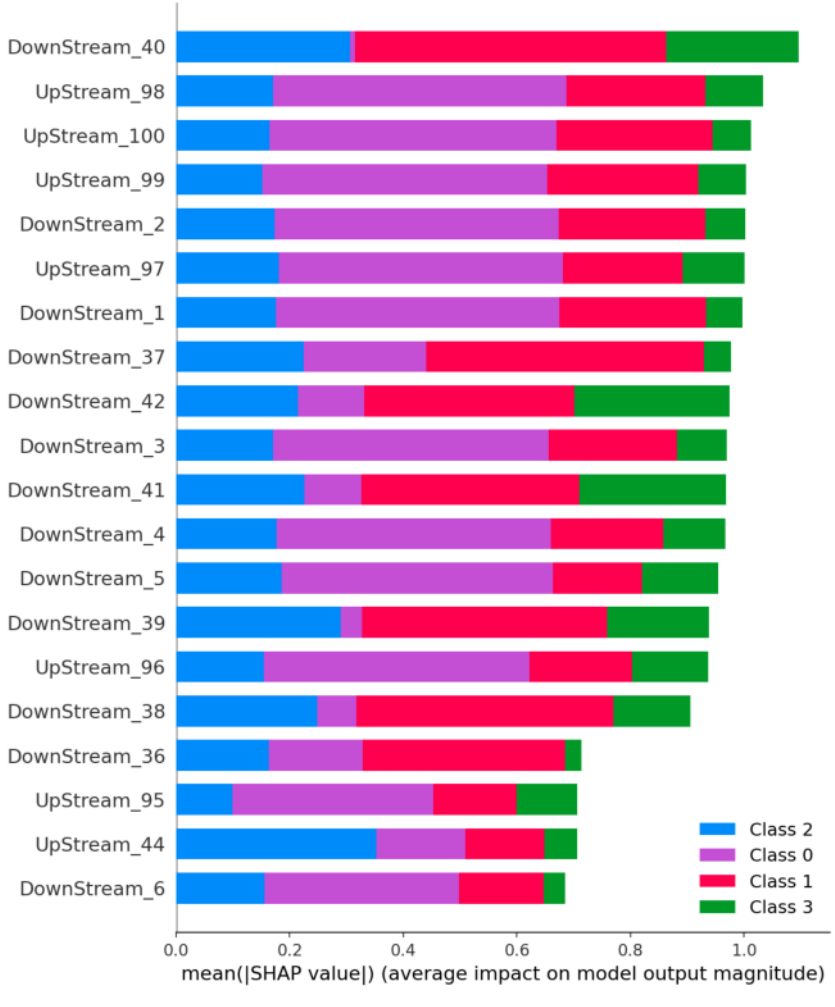


LOGISTIC_REGRESSION - SMOTE - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

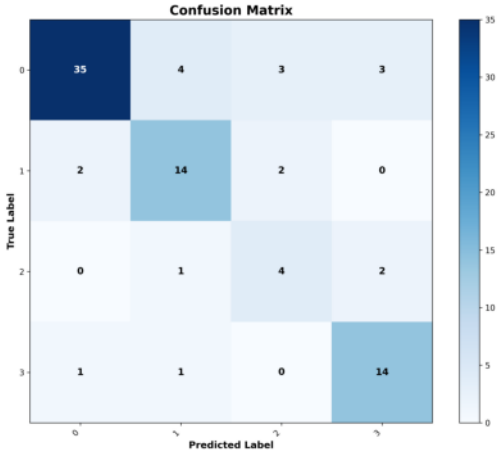


LOGISTIC_REGRESSION - RandomOverSampler Strategy

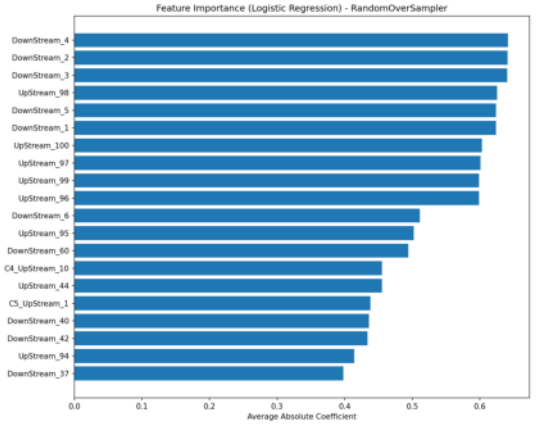
Performance Metrics

Accuracy	0.7791
Balanced Accuracy	0.7505
F1-Score (Macro)	0.7201
MSE	0.7326
MAE	0.3605
Log Loss	2.2229

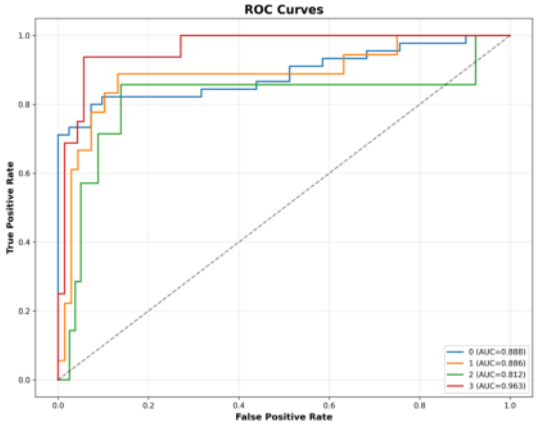
Confusion Matrix



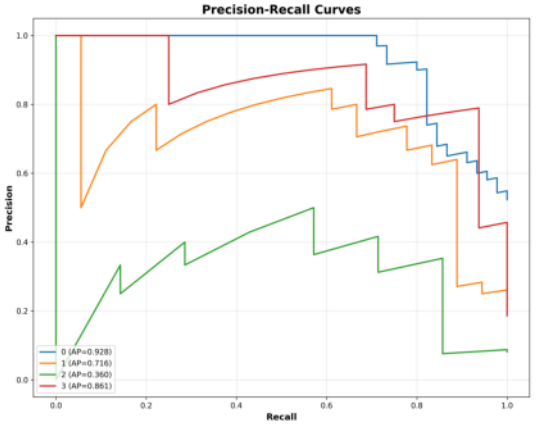
Feature Importance



ROC Curves

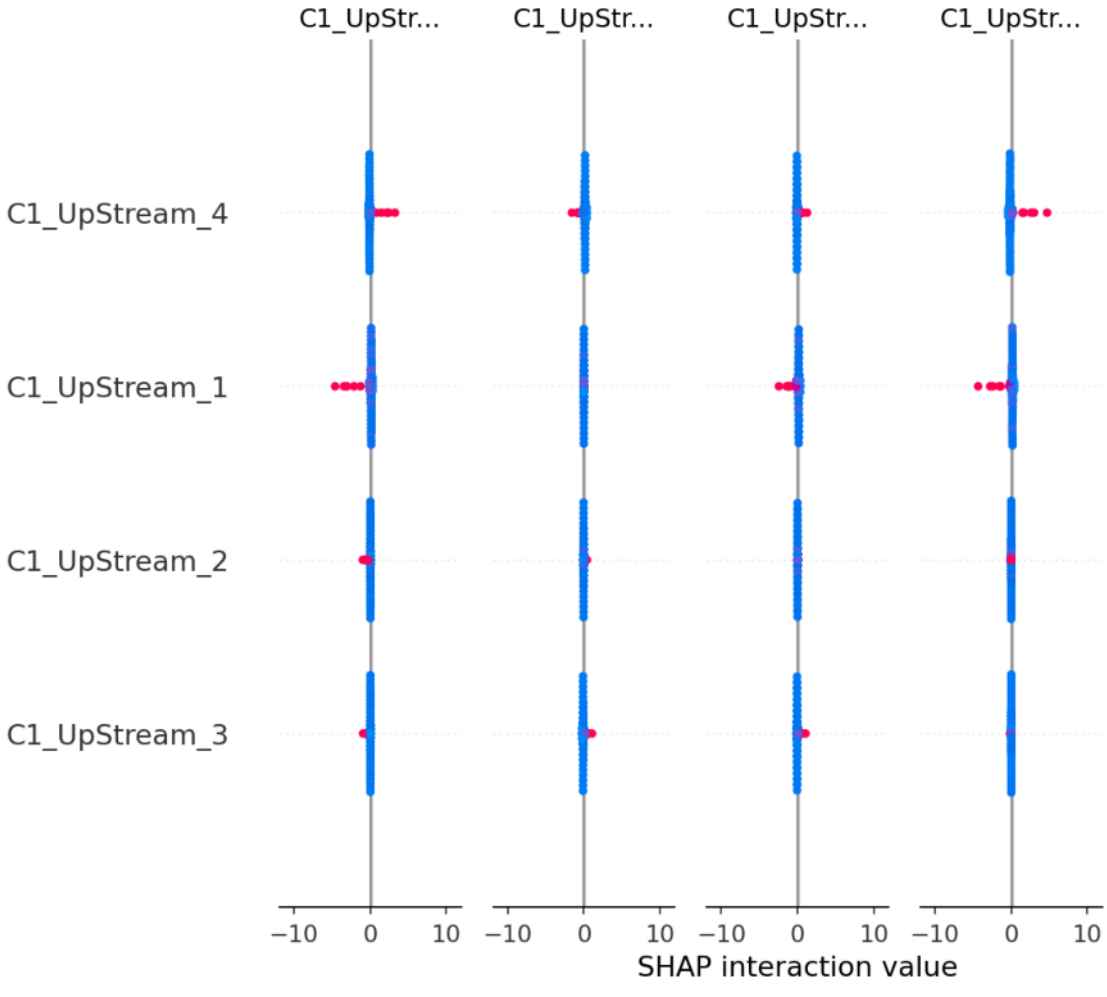


Precision-Recall Curves

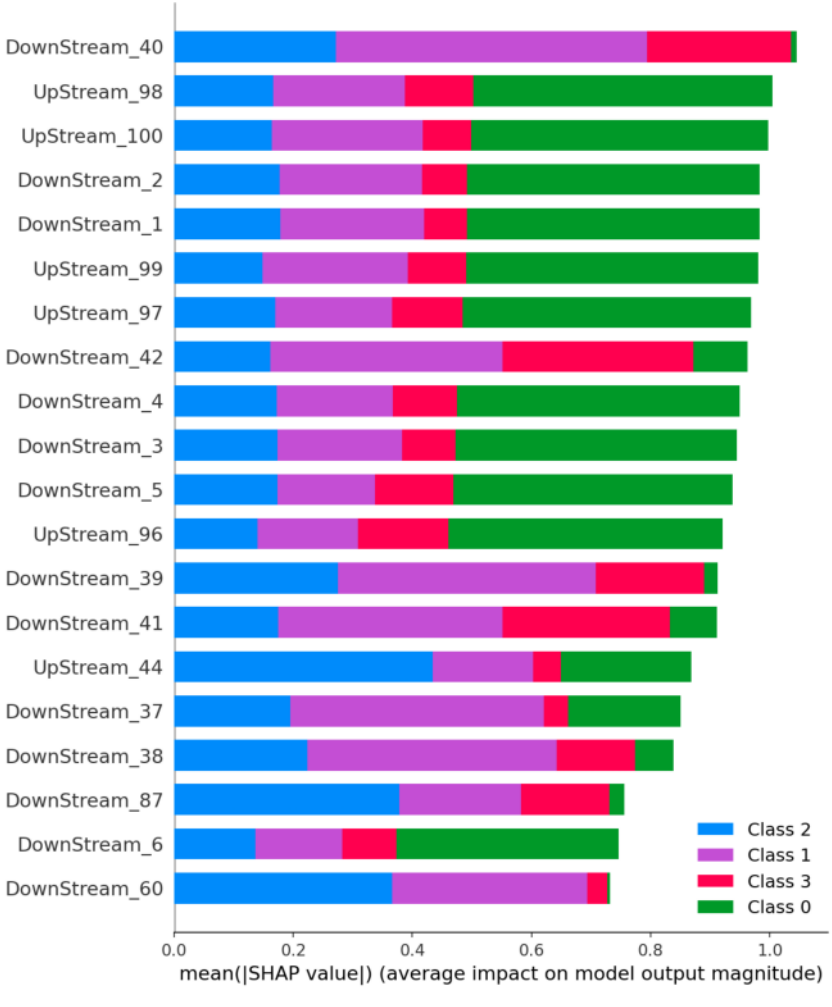


LOGISTIC_REGRESSION - RandomOverSampler - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

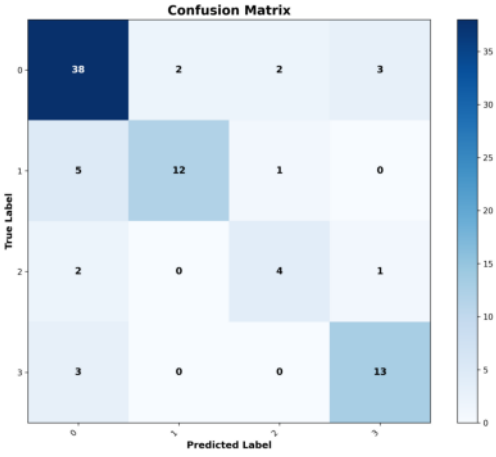


MLP - Original Strategy

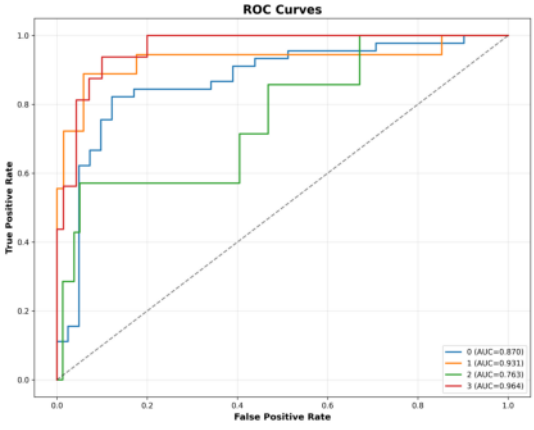
Performance Metrics

Accuracy	0.7791
Balanced Accuracy	0.7238
F1-Score (Macro)	0.7316
MSE	0.9186
MAE	0.4070
Log Loss	0.7914

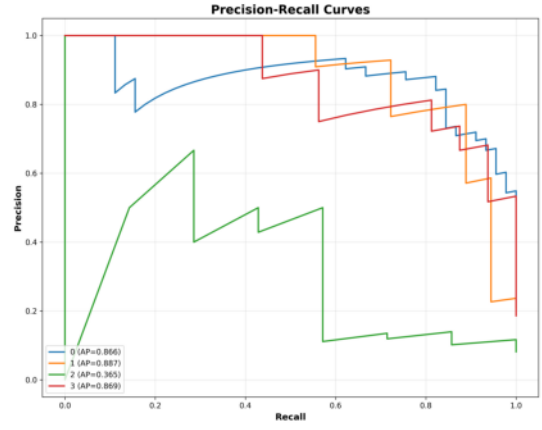
Confusion Matrix



ROC Curves



Precision-Recall Curves

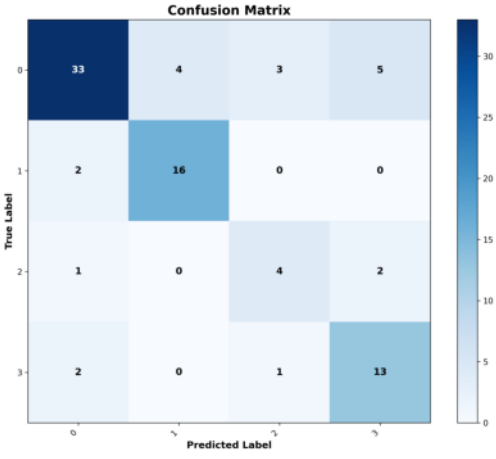


MLP - SMOTE Strategy

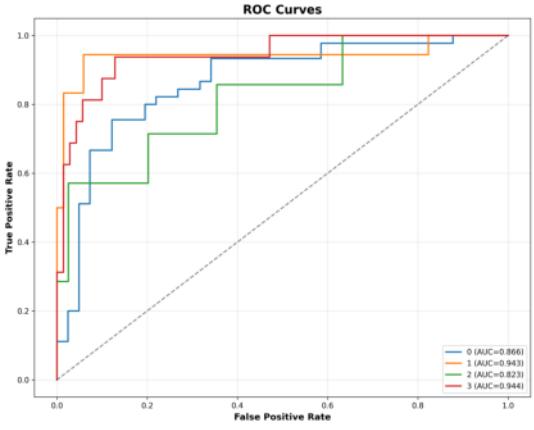
Performance Metrics

Accuracy	0.7674
Balanced Accuracy	0.7515
F1-Score (Macro)	0.7232
MSE	1.0233
MAE	0.4419
Log Loss	0.9889

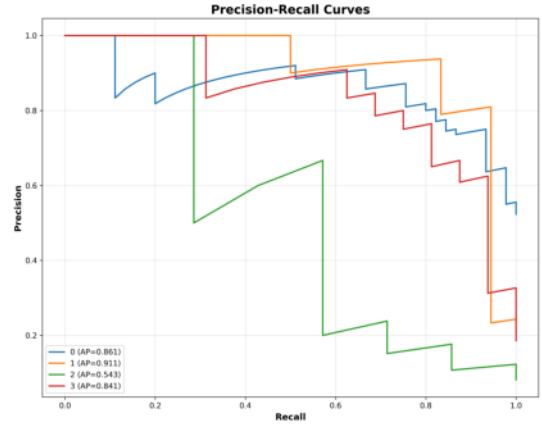
Confusion Matrix



ROC Curves



Precision-Recall Curves

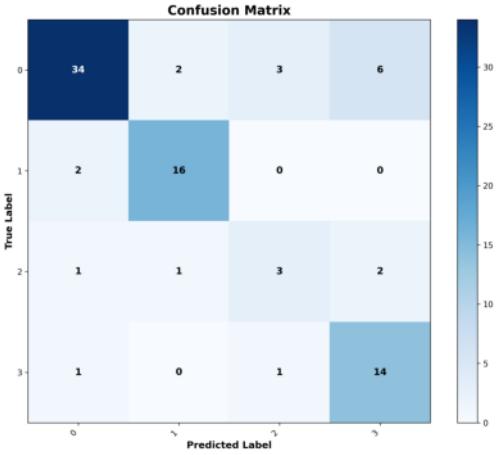


MLP - RandomOverSampler Strategy

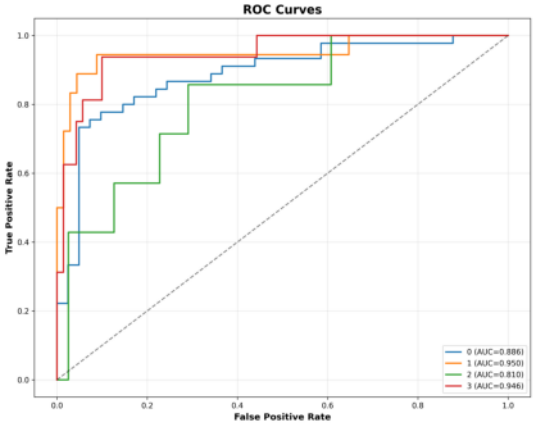
Performance Metrics

Accuracy	0.7791
Balanced Accuracy	0.7370
F1-Score (Macro)	0.7124
MSE	1.0116
MAE	0.4302
Log Loss	0.9160

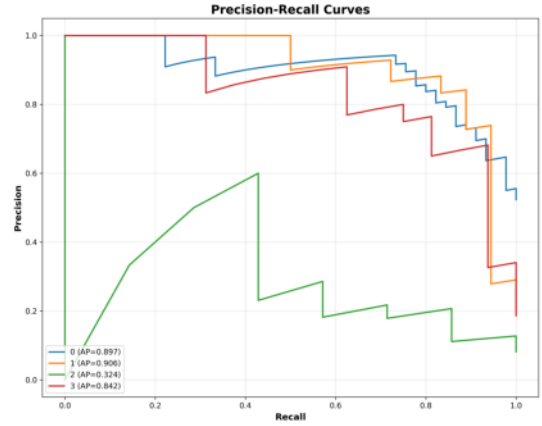
Confusion Matrix



ROC Curves



Precision-Recall Curves

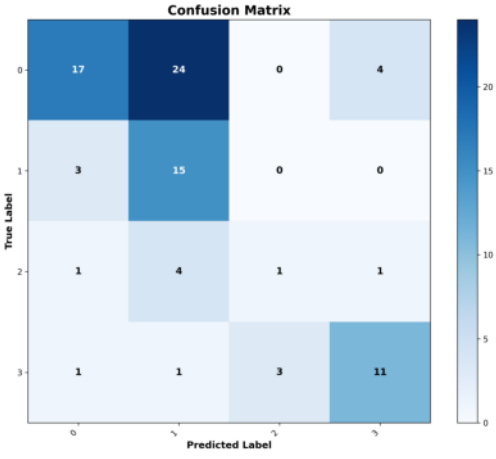


GNB - Original Strategy

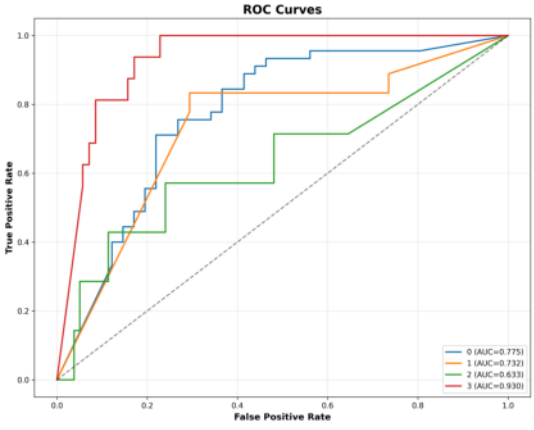
Performance Metrics

Accuracy	0.5116
Balanced Accuracy	0.5104
F1-Score (Macro)	0.4652
MSE	1.0233
MAE	0.6279
Log Loss	15.6952

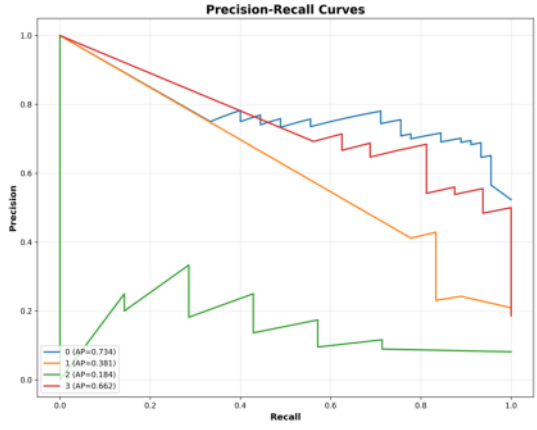
Confusion Matrix



ROC Curves



Precision-Recall Curves

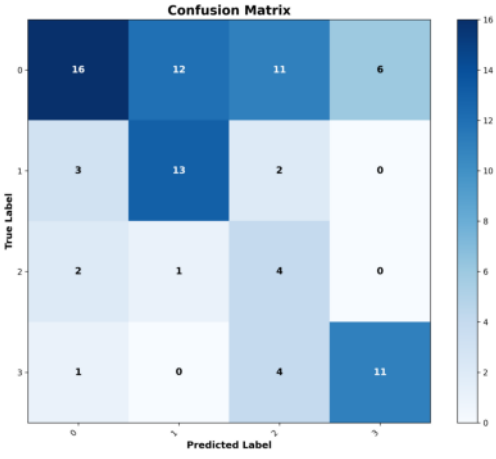


GNB - SMOTE Strategy

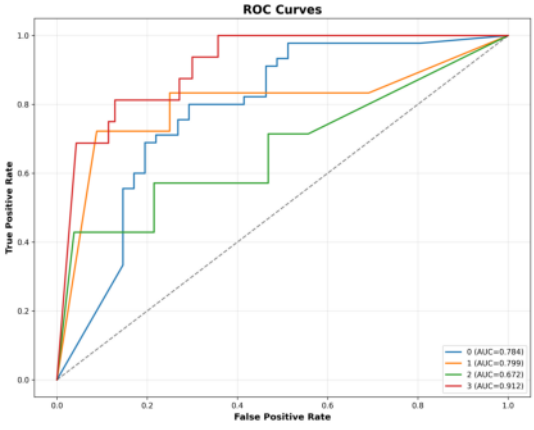
Performance Metrics

Accuracy	0.5116
Balanced Accuracy	0.5842
F1-Score (Macro)	0.5052
MSE	1.5930
MAE	0.8023
Log Loss	15.7342

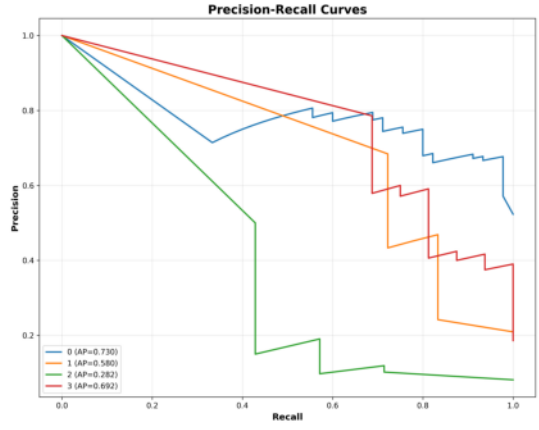
Confusion Matrix



ROC Curves



Precision-Recall Curves

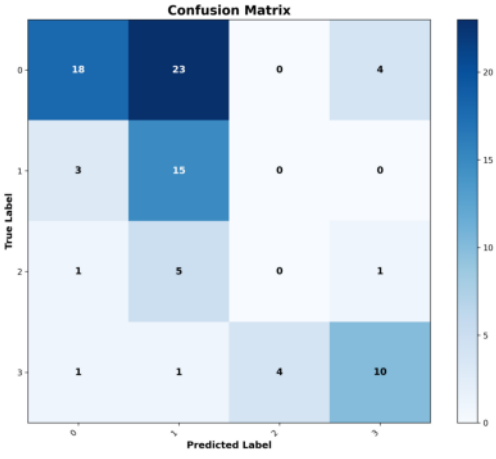


GNB - RandomOverSampler Strategy

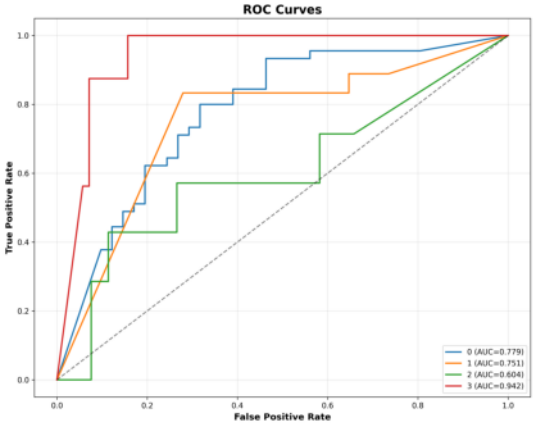
Performance Metrics

Accuracy	0.5000
Balanced Accuracy	0.4646
F1-Score (Macro)	0.4146
MSE	1.0349
MAE	0.6395
Log Loss	16.2135

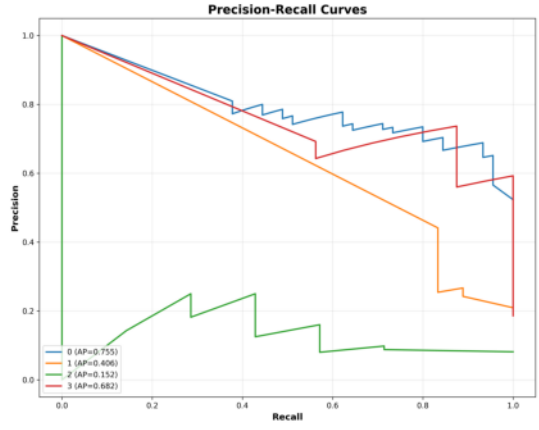
Confusion Matrix



ROC Curves



Precision-Recall Curves



Conclusions and Recommendations

KEY FINDINGS

Best Performing Model: LGBM with RandomOverSampler strategy
- Balanced Accuracy: 0.8508
- F1-Score (Macro): 0.8368
- Accuracy: 0.8488

OBSERVATIONS

1. RESAMPLING IMPACT

Different models respond differently to resampling strategies. SMOTE and RandomOverSampler can help with class imbalance, but may lead to overfitting in some cases.

2. MODEL COMPLEXITY

Tree-based models (Random Forest, XGBoost, LightGBM, Extra Trees) generally perform well on this dataset and provide interpretability through feature importance and SHAP values.

3. CLASS IMBALANCE

The dataset shows class imbalance. Balanced accuracy and per-class metrics provide better insights than overall accuracy alone.

4. FEATURE IMPORTANCE

SHAP values and feature importance plots reveal which features most strongly influence DNA artifact classification. Review these for domain validation.

RECOMMENDATIONS

✓ Use the best performing model for production deployment
✓ Monitor per-class performance, especially for minority classes
✓ Consider ensemble methods combining top performers
✓ Validate feature importance with domain experts
✓ Collect more data for underrepresented classes if possible
✓ Perform cross-validation for more robust performance estimates
✓ Test on independent validation set before deployment

NEXT STEPS

→ Hyperparameter tuning for top models
→ Cross-validation analysis
→ External validation dataset testing
→ Error analysis for misclassified samples
→ Feature engineering based on SHAP insights
→ Consider model calibration for probability estimates